Value Destruction and Recovery from Supply Chain Disruptions

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Problem definition: Prior literature documents significant effects of supply chain disruptions on firms' value with potential implications for the cost of capital. It also proposes long-term buyer-supplier relationships and multisourcing as key strategies for mitigating interruptions in operational flows. We study how sourcing strategies relate to market value resilience for firms experiencing disruptions, their customers and suppliers, and how market value resilience affects the cost of borrowing.

Methodology/results: We develop a fine-tuned large language model and apply it to historical press release data to identify disruption events at scale. To assess their impact, we link disruption events to supply network and stock return data. We introduce two measures, maximum value destruction (MVD) and time to recover market value (TTRMV), to quantify market-implied disruption severity and the speed of recovery. During 2003-2019, we find that disrupted firms, on average, experience an MVD of 2.2% and recover within 59 trading days. Value losses for suppliers and customers are less severe and are recovered faster. Longer buyer-supplier relationships, multisourcing, and customer diversification are associated with smaller value losses and faster recovery for disrupted firms, suppliers and customers. Among disrupted firms, longer buyer-supplier relationships result in a reduced cost of debt post-disruption.

Managerial implications: We present an AI-assisted workflow for disruption detection and establish a link between sourcing strategies and value resilience in supply chain disruptions. The results highlight the importance of long-term buyer-supplier relationships that increase market value resilience and lower the cost of debt.

Key words: supply chain disruption, supply network, relationship duration, stock returns, large language model, artificial intelligence

History: Mar. 2026

1. Introduction

Improving resilience to supply chain disruptions has been a central topic of operations management. Disruptions are consequential not only to material flows, but also to firms' value. The latter is important not only because it affects shareholders' wealth (Hendricks and Singhal 2003), but because it materially affects firms' operations through an increased cost of capital (Merton 1974, Campbell and Taksler 2003). This paper presents an AI-assisted method for detecting and quantifying the impact of supply chain disruptions on firms' valuations. We ask three research questions. First, following a disruption, what is the magnitude of market value destruction and how long does the market value recovery take? Second, how does the design of a firm's supply network, i.e., continuous long-term buyer-supplier relationships, multisourcing, or customer diversification, associate with firms' market value resilience during disruptions? Finally, do firms with higher market value resilience exhibit differences in post-disruption cost of capital?

The impact of supply chain disruptions on operational and financial health of firms varies. For example, the massive recalls of defective Takata airbags affected the supplier and multiple automakers, with Honda and Toyota experiencing valuation declines of 15-20% and roughly 5%, respectively (U.S. DOT 2015). Takata's equity sustained losses, its credit rating downgraded, and the firm ultimately filed for bankruptcy protection. In another episode, in February 2022, a contamination at a joint-venture facility operated by Western Digital and Kioxia (formerly Toshiba Memory) significantly impacted NAND flash memory production, and Western Digital's stock fell by more than 10% following the disclosure (Western Digital 2022). Yet the event was viewed as transitory, with market valuation recovering quickly and credit ratings maintained. These examples imply that (i) disruptions affect the financial state of firms and their supply chain partners, (ii) market valuations can provide useful information about disruption severity and firm resilience. The lens of financial markets in evaluating disruption impact is critical, because market valuation, as opposed to operational information, is readily available and accessible to investors and supply chain partners.

Our workflow starts with detecting disruption events from firm press releases, the historical record of which is collected from PR Newswire and Business Wire. We use a Llama 3 LLM (8B) fine-tuned on a human-labeled dataset of disruption disclosures to identify disruptions. Domain-specific fine-tuning substantially improves performance: the pretrained model achieves an AUC of 53.7%, while the fine-tuned model reaches 96.3%, comparable to ChatGPT-4o in the same task. Then we link identified events to focal firms via entity matching and map their upstream suppliers and downstream customers using FactSet Revere supply chain data. Daily stock returns for all public firms are obtained from CRSP.

Using stock return data, we construct two empirically grounded metrics to characterize the financial impact of supply chain disruptions: Maximum Value Destruction (MVD) and Time to Recover Market Value (TTRMV). Using an event-study framework based on cumulative raw and abnormal stock returns, MVD captures the maximum post-disruption loss in firm value, while TTRMV measures the time required for valuation to return to pre-disruption levels. Our baseline analysis uses cumulative buy and hold returns. Based on aggregated cumulative return paths, disrupted firms experience an MVD[1] of 2.2% and recover within 59 days, while suppliers and customers exhibit smaller MVD magnitude (0.70% and 0.78%) and faster recovery (41 and 42

[1] We use italicized $MVD/TTRMV$ to denote estimates and roman MVD/TTRMV for concepts.

days, respectively). Thus, the TTRMV and MVD metrics help downstream firms and investors assess risk exposure originating from upper-tier suppliers in a timely manner.

To inform firms on the effective management of the exposure, we examine how long-term buyer-supplier relationships, multisourcing, and customer diversification relate to market value resilience. We find that longer buyer-supplier relationships are associated with milder value losses and faster recovery. Specifically, a one-year increase in relationship length corresponds, on average, to 15.63 and 6.21 fewer days in $TTRMV$ for disrupted firms and suppliers respectively, and a 0.21% and 0.19% reduction in MVD magnitude for suppliers and customers. For customers, adding one supplier within the same industry as the disrupted supplier is associated with a 0.25% reduction in peak value loss and 7.34 fewer days to recover market value. For suppliers, adding one customer in the same industry as the disrupted customer corresponds to 2.8 fewer recovery days.

To analyze heterogeneity in disruptions and mitigate endogeneity concerns in the relationship between sourcing strategies and value resilience, we replicate the analysis on the set of disaster events. We find a stronger relationship between relationship length and MVD in the disaster-related sample (e.g., storms, earthquakes). The timing and intensity of disasters are plausibly exogenous conditional on firm fixed effects, which mitigates concerns that unobserved firm capabilities jointly determine sourcing strategies and disruption exposure. Finally, using a difference-in-differences analysis of disrupted firms, we find that longer buyer-supplier relationships result in a lower cost of debt in the post-disruption quarters.

Our paper makes the following empirical and methodological contributions. First, we introduce two empirically grounded metrics MVD and TTRMV that quantify the magnitude and duration of disruption impact on firms' value and are readily available from the public data. Second, in an event study framework, we establish the role of sourcing strategies in preserving firms' value. Third, we find that an increased market value resilience is associated with a lower cost of debt. Moreover, longer buyer-supplier relationships are causally related to a lower cost of debt, completing the circle between sourcing decisions, a firm's valuation and access to capital. Methodologically, we develop a fine-tuned LLM that detects supply chain disruptions in press releases in a scalable and replicable manner, addressing manual processing, a major bottleneck in previous research on supply chain disruptions. The fine-tuned LLM, together with the stock market data, quantifies firms' resilience, demonstrating the promise of AI-based text analytics in supply chain management (Cohen et al. 2025).

2. Literature and Hypotheses

Our work relates to the literature quantifying the impact of supply chain disruptions and the managerial approaches used to address them. In this section, we review these streams of literature and motivate our hypotheses regarding disruptions involving supply chain partners and the role of relationship and sourcing strategies.

2.1. Disruptions and Their Impact

The impact of disruptions in internal operations and supply chains has long been studied in the literature. The seminal work by Hendricks and Singhal (2003) documents a significant negative impact of disruption events on firms' market value. Subsequent research shows that disruption events are informative about firms' underlying operational weaknesses and are associated with poorer performance (Hendricks and Singhal 2005). Related work further demonstrates that such shocks are linked to adverse outcomes for economically connected firms, including reduced downstream output and declines in firm value along supply chain links (Barrot and Sauvagnat 2016). Focusing on stock returns, Cohen and Frazzini (2008) provide evidence that shocks to a buyer's stock return are associated with delayed return responses among its suppliers, an arbitrage opportunity that brought attention of financial markets to supply chain data. Our results on propagation of market value shocks to suppliers and customers align with Cohen and Frazzini (2008). Moreover, that propagation occurs with little lag, suggesting that supply chain linkage data is internalized by investors (Agarwal et al. 2017). However, more nuanced supply chain information, such as global sourcing data, may still not be fully priced by the market, as documented by Jain and Wu (2023).

In addition to the effect of disruptions, Carvalho et al. (2021) study microfoundations for disruption propagation. They argue that upstream firms, those supplying goods to disrupted firms, are affected because the disrupted firms serve as direct customers (i.e., sources of sales revenue). Therefore, when a focal firm is disrupted, its suppliers experience interruptions in purchases by that customer and see their own sales decline. On the other hand, downstream firms, those purchasing inputs from the disrupted firms, face a different mechanism. When a supplier is disrupted, its downstream customers experience a reduction in input availability. This supply scarcity drives up input prices, which in turn raises the cost of production for the downstream firms. These cost increases may be passed through as higher output prices, potentially leading to reduced demand and lower sales. While both mechanisms suggest adverse effects on connected firms, the downstream channel involves a longer and more indirect causal chain. Building on the findings of Cohen and Frazzini (2008), Hendricks et al. (2020) and Carvalho et al. (2021), we formulate the following hypothesis:

Hypothesis 1 *Following a supply chain disruption announcement, cumulative returns of upstream suppliers and downstream customers comove with those of the disrupted firm.*

2.2. Disruption Mitigation and Resilience

Significant research has focused on measuring and enhancing resilience to disruptions. Simchi-Levi et al. (2015) introduce Time to Recover (TTR) and Time to Survive (TTS) to identify critical suppliers and guide contingency planning, though TTR is often estimated via surveys and may be "unreliable" or "overly optimistic" (Simchi-Levi et al. 2015). In contrast, our metrics TTRMV and MVD are unambiguous, widely available, and applicable even to deep-tier suppliers, providing inputs for resilience optimization models (Simchi-Levi et al. 2015, Snoeck et al. 2019).

More recently, a number of papers studied best practices for supply chain resilience. Studying the role of sourcing strategies on recovery from interruptions of operational flows, Jain et al. (2022) provide evidence that maintaining long-term relationships and supplier centralization improve recovery speed from interruptions in the supply chain. Schmidt and Raman (2022) find that firms with credible internal control systems experience significantly smaller market value losses following disruptions. Investments in IT infrastructure, decentralized procurement, and artificial intelligence can also reduce damage from disruptions (Charoenwong et al. 2025, Han et al. 2025).

While the most direct approach to measuring supply chain resilience involves the use of operational data, obtaining such data is inherently difficult due to its proprietary and confidential nature. For example, Simchi-Levi et al. (2015) relied on a multi-year collaboration with Ford Motor Company and used masked versions of internal operational and supply chain data that are not publicly available. Kundu et al. (2024) interviewed 646 randomly selected small firms in Uganda to study how firms use redundancy strategies to enhance resilience. Mosayebi et al. (2024) utilized proprietary order-level field observations from a bus manufacturer to examine how the firm adapted to disruptions through design modifications, process changes, line balancing, and schedule adjustments.

As an alternative to primary operational data, Richards et al. (2025) relied on category-level prices of retail markets from Nielsen. In the context of pharmaceutical industry, Park et al. (2024) use the FDA drug shortage data to identify severity of disruptions. They propose *time to recover market share* as a measure of supply chain resilience, which is philosophically similar to ours. We follow Hendricks and Singhal (2003), Cohen and Frazzini (2008) and Han et al. (2025) and use financial data to quantify resilience to disruptions.

Buyer-supplier relationships data are prominent in information and decision support systems for investors, such as Bloomberg or FactSet. These information systems also track volume and length of the relationships. Generally, long-term buyer-supplier relationships are valuable to both parties because they unlock cost efficiencies for suppliers that are shared with customers (Kalwani and Narayandas 1995). Together with multisourcing, they promote sustained competition between suppliers (Chod et al. 2019). Arguments for the opposite effect of long-term relations include an excessive relational capital that creates rigidities and leads to suboptimal operational performance (Villena et al. 2011). In the context of market valuations, we posit that the net effect of long-term buyer-supplier relationships is in faster value recovery and milder value loss from disruptions, whether supply chain partners' or own. As for potential mechanisms, a longer relationship length may offer a higher priority in product allocation to a customer or business continuity to a supplier. In turn, suppliers or customers can reciprocate by being flexible or collaborating on disruption mitigation with their supply chain partners. Both mechanisms would result in a smaller value loss and faster recovery. Thus, we formulate the following:

Hypothesis 2 *Long-term buyer-supplier relationships are associated with a smaller MVD magnitude and shorter TTRMV for disrupted firms, as well as their suppliers and customers.*

Firms can multisource to manage their exposure to operational risks (Tomlin and Wang 2011). A classical example of this strategy is the case of Nokia facing a disrupted supply of radio frequency chips due to a fire at the Philips plant (Sheffi 2007). Beyond enhancing resilience, multisourcing can also foster supplier competition, incentivizing investments in recovery capabilities such as backup capacity and breakdown management procedures (Iyer et al. 2005). By sourcing from multiple suppliers, firms reduce their reliance on any single supplier, improving resilience to shortages, quality issues or losses of supplier capacity (Aydin et al. 2011). However, a recent study by Khanna et al. (2022) suggests that the effects of multisourcing may not be as robust as previously found in the OM literature. They posit that if a supplier knows that a customer multisources, they might de-prioritize that customer in resolving a disruption. Empirically they report that, in response to COVID-19 lockdowns, multisourcing firms are more likely to terminate their relationships with suppliers. Beyond operational impacts, multisourcing is visible to investors and can be reflected in market valuations of downstream supply-disrupted firms. Therefore, we propose the following hypothesis as the net effect of competing forces and test it against the null of no association between MVD, TTRMV and multisourcing:

Hypothesis 3 *Multisourcing is associated with a smaller MVD magnitude and shorter TTRMV for customers of disrupted firms.*

Firms can reduce risk by forming a portfolio of customers. For example, this strategy can help smooth demand variability and mitigate the bullwhip effect (Osadchiy et al. 2021). During the global COVID-19 pandemic, firms with a diversified portfolio of customers were shown to improve inventory efficiency amid supply chain disruptions (Ke et al. 2022). In contrast, a concentrated customer base can weaken a supplier's bargaining power and value capture, leading to lower profitability and heightened long-term financial risk (Saboo et al. 2017). Moreover, customer concentration increases a supplier's cost of equity and debt by elevating risk, particularly when key customers are difficult to replace (Dhaliwal et al. 2016). This leads to the following hypothesis:

Hypothesis 4 *Customer diversification is associated with a smaller MVD magnitude and shorter TTRMV for suppliers of disrupted firms.*

2.3. Market Value Resilience and Financing Outcomes

The preceding hypotheses focus on how sourcing strategies shape the magnitude of value destruction and the speed of recovery following disruptions. We now turn to whether differences in sourcing strategies translate into subsequent financing outcomes. In structural credit risk models, corporate debt is priced as a contingent claim on firm value (Merton 1974). Declines in firm value and increases in value volatility raise default risk and expected distress costs, leading to higher required yields. Empirically, equity volatility and downside risk are strongly associated with bond yield spreads and credit default swap premia (Campbell and Taksler 2003, Zhang et al. 2009). Supply chain disruptions, by generating sharp drawdowns and uncertainty about recovery, can therefore tighten financial constraints and increase borrowing costs. Conversely, firms with a high market-value resilience (e.g. employing long-term buyer-supplier relationships) signal lower distress risk and could have more favorable financing conditions in the post-disruption period. Accordingly, we posit:

Hypothesis 5 *Following a supply chain disruption, disrupted firms with longer buyer-supplier relationships experience lower cost of debt.*

From a managerial perspective, valuation instability can therefore tighten financial constraints and raise the cost of external capital. These considerations become particularly salient during supply chain disruptions, when uncertainty about its magnitude and duration increases and market values can adjust sharply. Preserving market value and recovering it quickly can reduce financing risk and help maintain access to debt markets precisely when liquidity is needed the most.

3. Data

We use three primary data sources: FactSet Revere, CRSP (Center for Research in Security Prices) daily security prices data, and firm press releases from Factiva. Our analysis focuses on supply chain disruptions announced between April 3, 2003, and December 31, 2019. This timeframe does not include the COVID-19 pandemic, which was a globally correlating disruption. The following subsections provide a detailed description of each data source used in the study.

3.1. FactSet Revere Supply Chain Relationship Data

To identify relationships between firms, we utilize FactSet Revere supply chain relationship dataset, focusing on the following key columns: a unique entity identifier generated by FactSet representing a firm in a buyer-supplier relationship, the start date and the end date of the relationship. The underlying data sources include government-mandated disclosures (forms 10-K, 10-Q, 8-K), press releases, investor presentations, company websites and social media. Since government disclosures are focused on the concentration of risk among key customers, we check whether the coverage over-represents larger downstream firms. E-Companion EC.2 demonstrates that customers and suppliers have highly overlapping size distributions, mitigating this concern.

3.2. CRSP Stock Price Data

Daily firm returns including dividends (RET) are obtained from CRSP, with value-weighted market returns ($VWRETD$) used to capture overall market performance. Firm identifiers and mappings to FactSet are based on 6-character CUSIPs and the WRDS FactSet-CRSP linking table.

3.3. Factiva Press Release Data

We collect firm press releases related to supply chain disruptions from Factiva. We use press releases distributed by *PR Newswire* and *Business Wire*, because they represent primary reports of the current events issued by firms or third parties (e.g., regulators). At the same time, we exclude *Dow Jones Newswire* because it primarily reports news articles and journalists' reports that represent a lagged and selective coverage of disruptions, focusing on disruptions that move financial markets. Our search for press releases spans the period from January 1, 2000, to December 31, 2019. While our supply chain relationship data are available only from April 3, 2003 onward, we keep disruptions identified between 2000 and 2003 to support the fine-tuning of our LLM. The search used two sets of keywords: variants of events, and variants of origins (Wu 2024). Variants of events refer to causes of disruptions (e.g., earthquake, collapse, etc.) or forms of disruptions (e.g., shutdown, halt, recalls, etc.). Variants of origins refer to where disruptions occur (e.g., production, materials, operations, etc.). A detailed description of the keywords and full search query is provided in Table 1. The query comprises two parts: the first searches for keywords in article headlines (using hd=), and the second applies the same keywords to the first 20 words of the main text (F20). Variants of events and origins are connected with NEAR5, requiring them to appear within five words of each other. The choice of the first 20 words and a 5-word proximity reflects a balance between capturing potentially relevant disruption information and avoiding excessive noise in news retrieval, as key information often appears early in articles and in close textual proximity. This structure ensures that the query effectively identifies relevant content while remaining focused. Because Factiva's duplicate removal relies on an undisclosed algorithm, we do not remove duplicates in the search

Table 1 Factiva press releases search strategy.

Variants of Events	problem break* backlog* damag* hurric* outbreak*	disrupt* malfun* obstruc* emergenc* landsli* contaminat*	interrup* fail* strike* fire* tsunami* hazard*	shutd* delay* lockout* explos* drought* blackout*	halt* shortag* incident* earthq* heatwav* sabotag*	stopp* bottlene* cris* flood* pandemic* collaps*	suspen* constrain* recall* storm* epidemic* evacuat*
Variants of Origins	produc* machin* downtim* parts logis* wareh*	manufact* equipm* throughp* inventor* transpor* carrier*	plant* capacit* procur* stock* ship* demand*	facilit* operat* sourc* goods* deliv* schedul*	assembl* workfl* suppl* distrib* freight* pipel*	fabricat* procedur* material* fulfil* cargo* terminal*	process* mainten* compon* order* transit* infrast*

Search Query in Factiva: [hd=[[EventVariant1 OR EventVariant2 OR ...] NEAR5 [OriginVariant1 OR OriginVariant2 OR ...]]] **OR** [[[EventVariant1 OR EventVariant2 OR ...] NEAR5 [OriginVariant1 OR OriginVariant2 OR ...]]/F20/]

process to ensure replicability. We retain full original text of press releases and metadata. The search was conducted on December 4, 2024 yielding 21,514 articles. To refine the dataset, we applied regular expressions to exclude articles unlikely to reflect supply chain disruptions. Specifically, we removed those containing terms such as “research”, “study”, “survey”, “market outlook”, or “breakthrough” within the first 20 words of the main text, as these are typically associated with industry or market research rather than reports of actual disruption events. We also excluded articles referencing sources such as “MarketsandMarkets” or “Understory”, which primarily publish syndicated market analysis. These filters reduced the sample to 12,450 articles.

4. Identification of Disruption Events

Manual labeling of disruption-related press releases, while valuable in early studies, is labor-intensive, potentially prone to errors, and difficult to replicate. Thus, automated processing of unstructured texts is increasingly used in supply chain studies. For instance, Wu (2024) constructs supply chain risk measures using a bag-of-words approach applied to firms’ 10-K filings, and Thiele et al. (2026) use it to study the effect of risk resolution. More recently, researchers have turned to LLMs to extract information and context that goes beyond keyword co-occurrence. Niu et al. (2025) use GPT-4 to characterize servitization features from 10-K reports, while Osadchiy et al. (2025) employ GPT-4o to extract customer-specific trade credit data.

General purpose LLMs, however, may not fully meet the needs of specialized research tasks. Fine-tuning becomes essential for achieving higher accuracy and domain relevance. For example, Siano (2025) fine-tunes a BERT (Bidirectional Encoder Representations from Transformers; Kenton and Toutanova 2019) model to predict short-window abnormal stock returns based solely on the textual content of earnings announcements. While BERT captures contextual meaning effectively, its limited context window (i.e., 512 tokens, approximately 380 words) is insufficient for processing lengthy financial documents. This limitation has motivated the use of generative LLMs, which support substantially longer context windows. For instance, both OpenAI’s GPT-4o and Meta’s

Llama models support context windows of up to 128,000 tokens (Grattafiori et al. 2024), enabling applications such as Sriraman et al. (2023), who fine-tune GPT models to link financing frictions with blockchain features, and Niu et al. (2024), who fine-tune a Llama3-Instruct model to classify supply chain related question-and-answer segments from earnings calls.

4.1. LLM Fine-Tuning

The LLM fine-tuning process consists of the following steps: *manual tagging*, *fine-tuning*, and *inference*. We start with pre-processing the texts by removing newline characters, carriage return characters, tabs, HTML tags, non-printable characters, and repeated punctuation and spaces from all press releases. For manual tagging, we randomly sample 10% of the 12,450 articles and manually tag them. While tagging, we broadly consider supply chain disruptions to include events that may affect a firm's operations or its ability to meet demand (e.g., storms, product recall, strike, outage, parts shortage). The labels for tagging are either '1' (when an article mentions a firm-specific supply chain disruption) or '0' (when an article does not mention any firm-specific supply chain disruption). After manual tagging, a random subset of 150 articles was reviewed to check the quality of tagging.

We fine-tune a Llama 3 8B pretrained model to build a text classifier that maps raw articles directly to binary labels, avoiding sensitivity to prompt phrasing inherent in instruct models, which are optimized for instruction-following and multi-domain tasks. Low-rank adaptation (LoRA; Hu et al. 2022) is applied to the 8-bit quantized model. The manually labeled dataset is split into training (70%), validation (15%), and test (15%) sets, with performance evaluated via AUC. Fine-tuning substantially improves performance, raising AUC from 53.7% to 96.3% on the test set (Figure 1). Further details are provided in E-Companion EC.1.

For comparison, Table 2 includes the classification performance of the GPT-4o on the test dataset. Our fine-tuned model performs comparably to GPT-4o in this task. We also report the inter-rater reliability using Cohen's kappa coefficient (Cohen 1960), which measures the level of agreement between two raters while accounting for agreement occurring by chance. Table 2 column "Cohen's kappa" presents the coefficients for three comparisons: human versus Llama3 before fine-tuning (0.105), human versus Llama3 after fine-tuning (0.764), and human versus GPT-4o (0.706). The fine-tuned model reports the highest Cohen's kappa. We saved the fine-tuned adapters for our next step, inference (text classification). Table EC.1 summarizes pairwise performance comparisons across the models.

4.1.1. Inference. We apply the same preprocessing to the remaining 90% of the press releases. Then we use the fine-tuned model to classify the remaining texts based on the arguments maximizing the logit function and obtain 2,538 press releases classified as "1". Within the manually tagged

Figure 1 **LLM performance on test data before and after fine-tuning.**

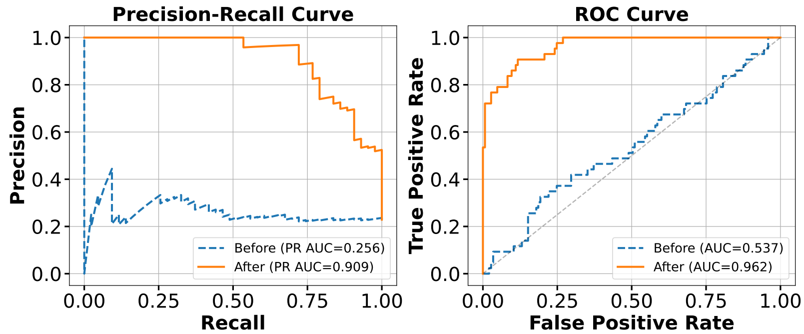


Table 2 **Confusion matrices before and after fine-tuning.**

Stage	True Class	Predicted 0	Predicted 1	Cohen's kappa
Before	Class 0	122	23	0.105
	Class 1	32	11	
After	Class 0	140	5	0.764
	Class 1	10	33	
GPT-4o	Class 0	137	8	0.706
	Class 1	11	32	

Table 3 **Examples of press releases and their classification by ChatGPT 4o vs. fine-tuned Llama3.**

Press Release Excerpt	GPT	FT	Human
Georgia Gulf Says Two Plants Remain Down Due to Hurricane Rita; Restart of Plants Dependent on Natural Gas, Feedstocks and Utilities Availability. ATLANTA - (BUSINESS WIRE) - Sept. 28, 2005 - Georgia Gulf Corporation (NYSE: GGC) said today that its Lake Charles, Louisiana facility, which produces vinyl chloride monomer (VCM) and its Plaquemine, Louisiana facility, which produces chlorine, caustic, VCM, vinyl resins, phenol and acetone, remain shutdown following Hurricane Rita until natural gas, feedstocks and utilities are restored ...	1	1	1
Giant Food Alerts Customers to Voluntary Recall of Giant Brand Garlic Bread and Garlic Spread. Products recalled due to unlisted milk products LANDOVER, Md., June 27 /PRNewswire-USNewswire/ – Giant Food has announced a voluntary recall of Giant brand garlic bread and garlic spread products sold in its Bakeshop as the items contain milk products which are not declared on the labels. Giant removed from its shelves all 16 ounce packages of Giant brand garlic bread and all 16 ounce packages of Giant brand garlic spread products. The following products are affected by this recall: – 16 oz. Giant Garlic Bread UPC ...	0	1	1
HireRight Employment Screening Report for Transportation Indicates Significance of Driver Shortage in an Improving Economy. IRVINE, Calif.–(BUSINESS WIRE)–May 14, 2013– Economic prospects for transportation companies are improving, but the struggle to hire experienced workers due to the driver shortage stands as the number one business challenge for a majority of companies. These findings from the Transportation Spotlight, a special report derived from the 2013 HireRight Employment Screening Benchmarking Report, highlights key developments in hiring and screening practices for the transportation industry. "Finding qualified drivers with acceptable safety records is job one for transportation companies in 2013," said Steven Spencer, vice president of transportation, HireRight. "The pinch is being felt more strongly than last year as a combination of regulatory, economic and social issues push drivers from the profession ..."	1	0	0

Note. GPT/FT/Human, classification by GPT-4o/Fine-Tuned Llama3 model/manual tagging respectively.

texts, there are 286 positive samples. In total, we have 2,824 articles identified as mentioning firm-specific supply chain disruptions by either human or the fine-tuned LLM. Table 3 illustrates how GPT-4o, our fine-tuned model, and human annotators assign differing labels to the same texts.

4.2. Entity Matching

To match press releases to CUSIPs and FactSet firm identifier, we rely on the following stratified approach: (i) ticker matching, (ii) firm name matching. The procedure is not trivial, because about 30% of press releases are issued by regulators, for example, the U.S. Consumer Product Safety Commission, and may mention more than one firm in the text. Moreover, the list of entities mentioned in a press release is not always available in Factiva's meta data. Because of that, we match ticker symbols or firm names directly to the text of press releases, and not to a separate list of tickers or firms. More details are available in E-Companion EC.3.

We provide descriptive statistics of the events in our data by industry, year, and type in Table 4. An event is defined as a unique pair consisting of a specific firm and an announcement date. Most of the events are in manufacturing (41.4%, 566), mining (21.1%, 288), and transportation, communications, electric, gas and sanitary services (18.9%, 259). Most years have consistent proportions of events except for 2005 and 2008, the years of the hurricanes Katrina and Ike, respectively. These patterns are consistent with Schmidt and Raman (2022) who use similar press release data and manual processing. Comparatively, our LLM-based process detected about 15% more events over 2003-2012, the common period in both studies. Based on the keywords present in the disruption announcements, they are categorized in supply-, logistics-, quality-, and disaster-related groups. We observe that the majority of events are in logistics and disaster groups.

5. Time to Recover Market Value and Maximum Value Destruction

Our measures of the impact of disruptions are based on financial returns. Before defining them formally, it is useful to set up key return constructs that they will be based on. We will use three subscripts: d for the disruption announcement date, i for the firm, and t for the number of trading days since the disruption announcement date. Time $t = 0$ represents the day of disruption announcement and $t = -1$ represents the previous trading day. Our analyses start from $t = -1$.

For firm i involved in a disruption announced on date d, let r_{dit} be the stock return on day t including dividends. For example, if firm i announces a disruption on date $d = 2026.01.20$, and $t = 3$, then r_{dit} represents firm $i's$ stock return on 2026.01.23 (i.e. on $d + t$). We measure all returns in basis points (i.e., bps): 1 bps = 0.01%. We construct cumulative returns (CR), specifically, the Buy-and-Hold CR (BAHCR) and the Buy-and-Sell CR (BASCR) between time t_1 and t_2 as:

$$BAHCR_{di}(t_1, t_2) = \prod_{k=t_1}^{t_2} (1 + r_{dik}) - 1, \quad \text{and} \quad BASCR_{di}(t_1, t_2) = \sum_{k=t_1}^{t_2} r_{dik}.$$

$BAHCR_{di}(t_1, t_2)$ and $BASCR_{di}(t_1, t_2)$ are similar when r_{dik} is small (Campbell et al. 1997), yet $BASCR_{di}(t_1, t_2)$ has a distinct interpretation: it represents the cumulative returns from the uniform

Table 4 Description of the disruptions.

Panel 1: Number of Events by Industry

SIC Division	Number of Events	%	SIC Codes
Agriculture, Forestry, And Fishing	3	0.2%	01-09
Construction	13	1.0%	15-17
Finance, Insurance, And Real Estate	61	4.5%	60-67
Manufacturing	566	41.4%	20-39
Mining	288	21.1%	10-14
Public Administration	16	1.2%	91-99
Retail Trade	50	3.7%	52-59
Services	66	4.8%	70-89
Transportation, Communications, Electric, Gas, And Sanitary Services	259	18.9%	40-49
Wholesale Trade	46	3.4%	50-51

Panel 2: Number of Events by Year

Year	Number of Events	%
2003	77	5.6%
2004	92	6.7%
2005	205	15.0%
2006	75	5.5%
2007	86	6.3%
2008	181	13.2%
2009	85	6.2%
2010	40	2.9%
2011	77	5.6%
2012	76	5.6%
2013	41	3.0%
2014	71	5.2%
2015	30	2.2%
2016	71	5.2%
2017	69	5.0%
2018	49	3.6%
2019	43	3.1%

Panel 3: Number of Events by Type

Supply ($N = 480$)	shortag*	supplier	procur*	sourcing	sourced	materials	component	parts	
Logistics ($N = 762$)	transpor*	ship	deliv*	freight	cargo	transit	logis*	carrier	delay
Quality ($N = 273$)	contaminat*	recall	hazard						
Disaster ($N = 727$)	heatwav* pandemic	explos* epidemic	earthq* outbreak	flood blackout	storm sabotag	hurric* collaps*	landslid* emergenc*	tsunami cris	drought fire

daily capital allocation strategy, i.e., investing an equal amount each day and aggregating daily profits at closing (O'Connell and Teo 2009).

To control for the market covariance in stock returns, following Hendricks and Singhal (2003), Hendricks et al. (2020), we calculate abnormal returns as:

$$AbRET_{dit} = r_{dit} - (\hat{\alpha}_{di} + \hat{\beta}_{di} R_{dt}) = r_{dit} - \hat{\alpha}_{di} - \hat{\beta}_{di} R_{dt}, \tag{1}$$

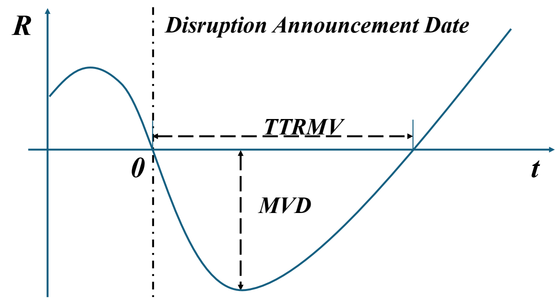
where R_{dt} is the market return on that day. Coefficients $\hat{\alpha}_{di}$ and $\hat{\beta}_{di}$ are estimated using the following regression model estimated on the window that ends 10 trading days before the announcement date (to avoid contamination from the event itself) and includes at least 40 and up to 200 trading days prior to that cutoff:

$$r_{dit} = \alpha_{di} + \beta_{di} R_{dt} + \epsilon_{dit}. \tag{2}$$

We consider two types of the cumulative abnormal returns (CAR): buy-and-hold CAR ($AbBAHCR$), and buy-and-sell CAR ($AbBASCR$). Following Hirshleifer et al. (2009), Barrot and Sauvagnat (2016), for firm i and disruption date d, we define $AbBAHCR$ and $AbBASCR$ from day t_1 to t_2 as:

$$AbBAHCR_{di}(t_1, t_2) = \prod_{k=t_1}^{t_2} (1 + r_{dik}) - \prod_{k=t_1}^{t_2} (1 + \hat{\alpha}_{di} + \hat{\beta}_{di} R_{dk}), \text{ and } AbBASCR_{di}(t_1, t_2) = \sum_{k=t_1}^{t_2} AbRET_{dik}.$$

Figure 2 Construction of TTRMV (time to recover market value) and MVD (maximum value destruction) measures.



5.1. Definitions

To quantify the impact of the disruption announced at d on the value of firm i and its recovery, we define Time to Recover Market Value (TTRMV), and Maximum Value Destruction (MVD) as:

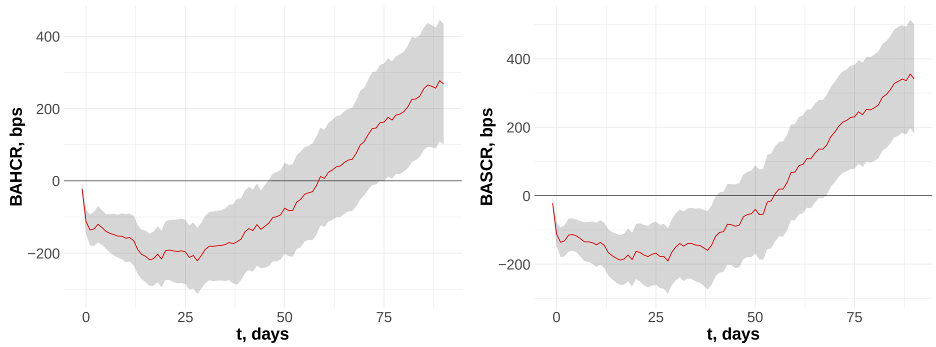
$$TTRMV_{di} = \min\{t \geq -1: \; R_{dit} \geq 0\},$$
$$MVD_{di} = \min_{t}\{R_{dit}: \; t \in [-1, TTRMV_{di}]\}.$$

Above, R_{dit} can be $BAHCR_{di}(-1,t)$, $BASCR_{di}(-1,t)$, $AbBAHCR_{di}(-1,t)$, or $AbBASCR_{di}(-1,t)$. Recovery is defined as the first date on which cumulative (abnormal) returns since the disruption announcement reach the pre-disruption market value benchmark (i.e., $R_{dit} \geq 0$). We visualize the $TTRMV$ and MVD metrics in Figure 2.

When firms experience supply chain disruptions, their stock prices typically show immediate negative returns, as investors update their expectations regarding future earnings, operational stability, and recovery prospects. During the recovery period, the disrupted firm's market value remains undervalued relative to its pre-disruption level. The duration of this period is captured by $TTRMV$. The period of undervaluation can undermine investor confidence, damage the firm's reputation, and even hinder access to external financing, such as bank loans. In that sense, TTRMV can be interpreted as a "loss of time" for the firm's shareholders during which firm value remains below its pre-disruption benchmark.

The MVD measure is complementary to TTRMV and represents the second degree of freedom in a relationship between a disruption and recovery: the depth of value destruction. Our measures are distinct from the well known short-term abnormal returns around periods of supply chain glitches (Hendricks and Singhal 2003) or fixed term returns (Hendricks et al. 2020). Both $TTRMV$ and MVD are computed over longer periods of time and capture longer-horizon market responses associated with disruption events. In §5.3, we also analyze the connection between $TTRMV/MVD$ and short-term measures of Hendricks and Singhal (2003), specifically, whether $TTRMV/MVD$

Figure 3 **Cumulative returns of disrupted firms (in basis points, the shaded areas denote 90% empirical confidence intervals).**



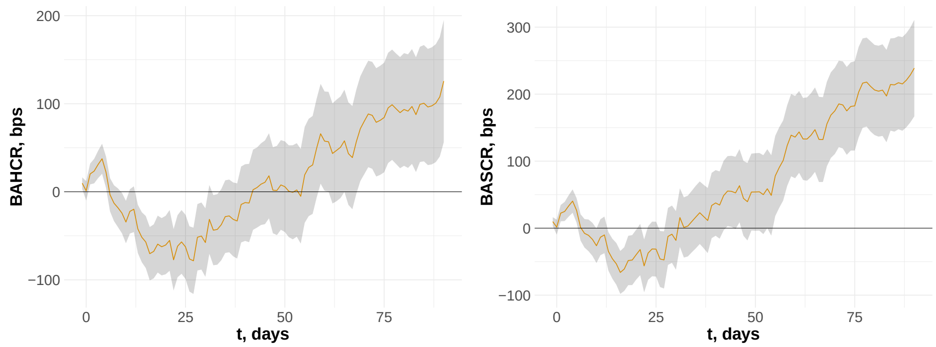
can be predicted by short term returns. $TTRMV/MVD$ also differ from the text-based supply chain risk measure ($SCRisk$) proposed by Wu (2024) in two key respects. First, $TTRMV/MVD$ are event-based measures constructed around specific supply chain disruption announcements, whereas $SCRisk$ reflects firms' general supply chain risk awareness as voiced in routine earnings conference calls. Second, $TTRMV/MVD$ not only measure market value risk, but also quantify the subsequent value recovery dynamics and the severity of realized value destruction, enabling a market-based assessment of supply chain resilience.

In our main analyses, we use raw $BAHCR$ returns to characterize TTRMV and MVD, while controlling for heterogeneity in the market performance. We choose raw returns because they reflect shareholders' wealth, are related to firms' cost of capital, and executive compensation. In that sense, $TTRMV$ and MVD based on raw returns have a clear interpretation of the "loss of time" and "loss of wealth" due to a disruption in current market conditions. Abnormal returns are interpreted differently: zero abnormal return some time after a disruption means that the company not only recovered to its pre-disruption value, but also caught up to the market. The latter is a much higher bar that may take a long time to clear, increasing the likelihood of confounding events (e.g., other disruptions). With that caveat, below we supplement our analysis with abnormal returns, adjusted for the overall market performance. With the market performance factored out, recovery trajectories show more noise, with some firms not being able to recover at all (i.e., have $TTRMV$ extend beyond the period of study), a phenomenon first documented by Hendricks and Singhal (2003).

5.2. Model Free Evidence

In this section, we demonstrate the impact of disruption events on disrupted firms, their customers and suppliers. For disrupted firms, both $BAHCR$ and $BASCR$ become significantly negative following announcements of disruptions, as shown in Figure 3. The MVD implied by the aggregated $BAHCR$ trajectory is -221.43 basis points, corresponding to a 2.2% loss, while the time to recover

Figure 4 **Cumulative returns of customers (in basis points, the shaded areas denote 90% empirical confidence intervals).**



market value ($TTRMV$) is 59 trading days. A small fraction of firms (5.35%) had persistent market value losses that were not recovered during our study period. These observations are included in our analysis and reflected in the estimates.

Abnormal returns show statistically significant negative impacts as well, see Figure EC.2. To benchmark our estimates against prior work, we align post-disruption horizons with those used in the literature. At 40 trading days, the horizon in Barrot and Sauvagnat (2016), we estimate an average $AbBASCR$ of -2.03%, comparable to their reported -2.49%. At 20 trading days, matching the window in Hendricks et al. (2020), we find an average $AbBASCR$ of -1.48%, which is smaller in magnitude than the -7.19% documented for firms affected by the Great Tohoku Earthquake. The results show that disruption announcements negatively affect firm market value, with persistently negative $AbBAHCR$ indicating lasting impact.

Following disruption announcements by focal firms, customer $BAHCR$ and $BASCR$ exhibit a short-lived small increase followed by a statistically significant decline. As illustrated in Figure 4, the implied MVD for customers is -78.37 basis points (0.78% loss), and the $TTRMV$ is 42 trading days. Both $AbBAHCR$ and $AbBASCR$ reveal statistically significant negative abnormal returns for customers following disruption announcements of focal firms, see Figure EC.3. However, their magnitude is smaller compared to the disrupted firms. Similarly to the disrupted firms, the customers' buy-and-hold abnormal returns do not recover within the post-event window.

Supplier stock returns exhibit a short-lived small increase followed by a statistically significant decline over a longer period, Figure 5. The corresponding MVD for suppliers is -70.45 basis points (0.70% loss), and the $TTRMV$ is 41 trading days. Despite a smaller MVD magnitude and shorter $TTRMV$ for suppliers, their abnormal returns appear to be larger in magnitude compared to customers; see Figure EC.4. Hendricks et al. (2020) similarly report larger abnormal returns upstream. Discrepancies between MVD and $TTRMV$ based on raw returns may reflect stronger upstream market correlations, due to factors like more durable output (Gomes et al. 2009), higher systematic demand risk (Osadchiy et al. 2016), or greater exposure to aggregate productivity (Gofman et al.

Figure 5 **Cumulative returns of suppliers (in basis points, the shaded areas denote 90% empirical confidence intervals).**

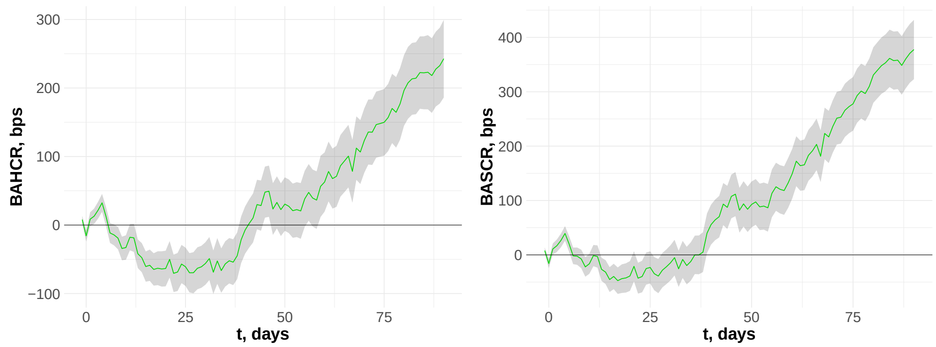


Table 5 **Empirical estimates of $TTRMV$ and MVD by supply chain position and disruption type.**

	Focal		*Customers*		*Suppliers*	
Disruption Type	*TTRMV*, days	*MVD*, bps	*TTRMV*, days	*MVD*, bps	*TTRMV*, days	*MVD*, bps
All	59	-221.43	42	-78.37	41	-70.45
Supply	120	-465.16	18	-44.36	38	-114.77
Logistics	85	-358.63	156	-193.81	45	-120.63
Quality	9	-164.92	2	-22.83	4	-44.61
Disaster	83	-351.92	155	-268.04	126	-285.42

2020). Table 5 summarizes the impact of disruptions on $TTRMV$ and MVD for disrupted firms, their suppliers and customers, and provides a breakdown by disruption type. We observe that disasters and supply disruptions have a stronger impact on market value. Note that the $TTRMV$ and MVD reported in Figure 3-5 and Table 5 are computed from the aggregated mean cumulative return path across disruption events and therefore represent descriptive summaries of average return dynamics. In §6-7, we investigate the role of sourcing strategies for mitigation of the impact of disruption on market value.

5.3. Connection to Short-Term Returns

To test whether $TTRMV$ and MVD can be predicted by short term returns, we regress them on the respective 3-day cumulative abnormal returns, as in Hendricks and Singhal (2003). Table EC.2 shows that even though 3-day returns are statistically predictive of $TTRMV$, particularly when it is based on abnormal returns, the explanatory power is low, with R^2 near zero. In contrast, the prediction of MVD by the 3-day returns is more reasonable, with R^2 ranging from 0.13 to 0.34 across disrupted firms, their customers, and suppliers. Specifically, for every 1% increase in 3-day returns, MVD increases by approximately 0.64% to 1.01% for disrupted firms, 0.53% to 0.76% for customers, and 0.52% to 0.68% for suppliers. These results suggest that short-term market reactions provide some signal of value destruction but offer limited information into recovery dynamics.

6. Sourcing Strategies and Value Resilience

Firms can use a variety of strategies to address disruptions. Grounded in the literature, this section explores the role of long-term buyer–supplier relationships, multisourcing, the availability of alternative sourcing options, and customer diversification in shaping firms' disruption outcomes.

6.1. Disrupted Firms

For each firm and disruption, we measure the actual average duration of relationships with supply chain partners, i.e. suppliers and buyers, as:

$$\overline{RL}_{di} = \frac{\sum\limits_{j \in SPLR_{di}} RL_{dij} + \sum\limits_{j \in CUST_{di}} RL_{dij}}{|SPLR_{di}| + |CUST_{di}|}.$$

Let the disrupted firm i's supplier (customer) base be $SPLR_{di}$ ($CUST_{di}$) at d, then the cardinality $|SPLR_{di}|$ ($|CUST_{di}|$) represents the number of active suppliers (customers) of firm i on disruption announcement date d. The relationship length, measured in years, between firm i and its supplier (customer) j, at disruption announcement time d is represented by RL_{dij}.

Associations between supply chain relationships and the returns of firm i can be estimated by:

$$BAHCR_{dit} = \alpha_i + \delta_{yq} + \beta_1 t + \beta_2 t^2 + \beta_3 \overline{RL}_{di} + \beta \mathbf{M}_{di} + \epsilon_{i,yq,t}. \tag{3}$$

The fixed effect α_i controls for time invariant firm-level unobservables, while δ_{yq} controls for the average market performance in year-quarter yq of disruption announcement date and other common time shocks. We estimate the relationship on the balanced panel of all disrupted firms for $t \in [0, 60]$, and include the quadratic term to capture the non-linear relationship between returns and time since disruption, as suggested by the "U-shaped" trajectory on Figure 3. $\mathbf{M}_{di}$ include firm-level controls: market capitalization ($MCAP$) in the latest quarter before disruption announcement, stock liquidity ($LIQ = [\sum\limits_{d-252}^{d-1} VOL_t / |RET_t|]/N$, VOL being daily trade volume, $|RET|$ being the absolute value of the daily return, N being the number of days of non-zero RET in the 252-trading-day window) measured in the latest 252 trading days before disruption announcement, and stock volatility ($VLT = \sigma_{d-252}^{d-1}(RET_t)$) in the latest 252 trading days before disruption announcement.

Next, we interact each variable of interest with t and t^2 to examine how supply chain relationships are related to the evolution of outcomes following disruption events:

$$BAHCR_{dit} = \alpha_i + \delta_{yq} + \beta_1 t + \beta_2 t^2 + \beta_3 \overline{RL}_{di} + \beta_4 t \times \overline{RL}_{di} + \beta_5 t^2 \times \overline{RL}_{di} + \beta \mathbf{M}_{di} + \epsilon_{i,yq,t}. \tag{4}$$

Specifications (3)-(4) are associative. Indeed, $\overline{RL}$ is a choice made by firms, which could be guided by observable or latent variables. Learning from prior disruptions is one example of such dependence. Analyzing the relationship over time, we observe no increased importance of $\overline{RL}$ or other sourcing strategies. Another channel is anticipation of future disruptions, for example the presence of an unreliable supplier may increase the likelihood of multisourcing. For that, in §7.4, we turn to the analysis of the subsample of disaster-related disruptions that conditional on the firm's fixed effect are exogenous to the firm's strategic decisions.

6.2. Customers (supply-disrupted) and Suppliers (demand-disrupted)

To examine how supply chain relationships are associated with the returns of customers and suppliers following disruption announcements, we estimate the following regression:

$$BAHCR_{dit} = \alpha_i + \delta_{yq} + \beta_0 focal\_BAHCR_{dit} + \beta_1 t + \beta_2 t^2 + \beta_3 RL_{di} + \beta_4 \ln(SDS_{di}+1) + \beta\mathbf{M}_{di} + \epsilon_{i,yq,t}. \tag{5}$$

In the above equation, δ_i and δ_{yq} represent customer (supplier), and disruption announcement year-quarter FEs respectively. Note that i is not disrupted, but its supplier (customer) is disrupted, announced at d. Thus, we can denote the disrupted supplier (customer) as $focal_{di}$. $focal\_BAHCR_{dit}$ is the $BAHCR$ of $focal_{di}$ on day t. RL_{di} is the relationship length, measured in years, between customer (supplier) i and the disrupted firm $focal_{di}$ at d. SDS_{di} represents "sourcing-distribution strategy". When i is a customer, $SDS_{di} = MS_{di}$, where MS_{di} measures customer i's multisourcing level. Let customer i's supplier base be $SPLR_{di}$ at d, and for firm $j \in SPLR_{di}$, $SIC(j)$ represents its 4-digit SIC, then we define MS_{di} as:

$$MS_{di} = \sum_{j \in SPLR_{di}, j \neq focal_{di}} \mathcal{I}_{\{SIC(j)=SIC(focal_{di})\}}.$$

MS_{di} is the number of alternative suppliers in the same SIC sector as the disrupted supplier $focal_{di}$.

When i is a supplier, $SDS_{di} = CD_{di}$, which measures customer diversification of i. CD_{di} is the number of alternative customers that are in the same SIC sector as the disrupted customer $focal_{di}$. Let supplier $i's$ customer base be $CUST_{di}$ at d, then we define CD_{di} as:

$$CD_{di} = \sum_{j \in CUST_{di}, j \neq focal_{di}} \mathcal{I}_{\{SIC(j)=SIC(focal_{di})\}}.$$

Since both MS_{di} and CD_{di} can be zero, we use $\ln(SDS_{di}+1)$ in regression (5). The indicator $\mathcal{I}_{\{SIC(j)=SIC(focal_{di})\}}$ equals 1 when "$SIC(j) = SIC(focal_{di})$" and 0 otherwise.

In specification (6), we interact each operational variable with t and t^2:

$$\begin{aligned} BAHCR_{dit} = {} & \alpha_i + \delta_{yq} + \beta_0 focal\_BAHCR_{dit} + \beta_1 t + \beta_2 t^2 + \beta_3 RL_{di} + \beta_4 \ln(SDS_{di}+1) + \beta_5 t \times RL_{di} \\ & + \beta_6 t \times \ln(SDS_{di}+1) + \beta_7 t^2 \times RL_{di} + \beta_8 t^2 \times \ln(SDS_{di}+1) + \beta\mathbf{M}_{di} + \epsilon_{i,yq,t}. \end{aligned} \tag{6}$$

We use the same set of control variables for customers and suppliers: $MCAP$, LIQ and VLT. In an additional analysis, we include supply chain alternatives density (SCAD) as a control variable. For customers/suppliers of disrupted firms, we include upstream/downstream SCAD (i.e., the number of unlinked firms with the same 4-digit SIC code as the disrupted suppliers/customers).

Table 6 Summary statistics for the sourcing strategy, valuation, and control variables.

Panel 1: Disrupted Firms

	Mean	Std	Min	Q25	Median	Q75	Max	Full Sample	SIC 13	SIC 28
$\overline{RL}$	1.46	0.77	0.06	0.91	1.37	1.83	4.87			
$\ln MB$	0.71	0.83	-2.23	0.18	0.62	1.15	3.42			
$\ln MCAP$	8.66	1.99	3.72	7.21	8.83	10.13	12.50			
$\ln LIQ$	19.31	1.69	14.84	18.25	19.37	20.56	23.65			
$\ln VLT$	-3.90	0.49	-4.76	-4.26	-3.96	-3.59	-2.55			
Number of Events								818		
Number of Disrupted Firms								443		
Number of Observations								50601		

Panel 2: Customers of Disrupted Firms

	Mean	Std	Min	Q25	Median	Q75	Max	Full Sample	SIC 13	SIC 28
MS	2.35	5.93	0.00	0.00	0.00	2.00	59.00			
RL	2.39	2.55	0.03	0.67	1.53	3.13	16.65			
$\ln MB$	0.80	0.85	-2.93	0.23	0.70	1.26	3.60			
$\ln MCAP$	8.41	2.34	2.90	6.71	8.37	10.25	12.79			
$\ln LIQ$	19.02	2.06	13.74	17.64	19.05	20.62	23.52			
$\ln VLT$	-3.90	0.52	-4.80	-4.30	-3.95	-3.56	-2.47			
Number of Events								852		
Number of Disrupted Firms								465		
Number of Customers								1624		
Number of Observations								312474		

Panel 3: Suppliers of Disrupted Firms

	Mean	Std	Min	Q25	Median	Q75	Max	Full Sample	SIC 13	SIC 28
CD	2.68	5.06	0.00	0.00	1.00	3.00	75.00			
RL	2.39	2.57	0.02	0.66	1.46	3.22	16.72			
$\ln MB$	0.82	0.83	-2.46	0.29	0.75	1.25	3.55			
$\ln MCAP$	7.08	2.18	2.39	5.57	6.98	8.51	12.46			
$\ln LIQ$	18.01	2.04	13.23	16.66	17.93	19.39	23.37			
$\ln VLT$	-3.69	0.50	-4.71	-4.04	-3.70	-3.36	-2.45			
Number of Events								810		
Number of Disrupted Firms								450		
Number of Suppliers								2581		
Number of Observations								587827		

6.3. Summary Statistics

Table 6 summarizes the data and variables used in the analyses. There are 818 events where disrupted firms have at least one supplier and at least one customer. This set of events is used to study an association between risk mitigating strategies and TTRMV and MVD. A median disrupted firm has a relationship length of 1.37 years.

To examine the resilience of supply-disrupted customers (Panel 2), we retain disrupted firms with at least one customer, and require that each customer has SIC code data for its suppliers, enabling computation of MS. This yields 852 events. For demand-disrupted suppliers (Panel 3),

Table 7 Association between sourcing strategies and post disruption returns for disrupted firms. Here and throughout the paper, standard errors are clustered by firm.

	(1) BAHCR (3)	(2) BAHCR (4)
t	-8.14	-23.24^{*}
	(5.83)	(9.92)
t^2	0.20^{**}	0.28^{*}
	(0.06)	(0.13)
RL	56.37	-175.72
	(227.26)	(256.61)
$t \times \overline{RL}$		10.08^{**}
		(3.72)
$t^2 \times \overline{RL}$		-0.06
		(0.10)
N	43559	43559
Firm FE	381	381
Year-Quarter FE	67	67
Controls	✓	✓
R^2	0.52	0.52

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$

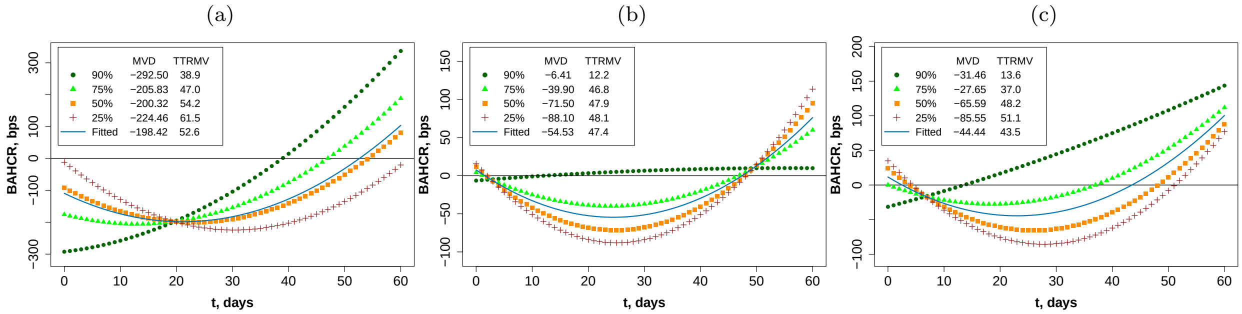
we retain disrupted firms with at least one supplier, and require that each supplier has SIC code data for its customers, enabling computation of CD. This yields 810 events.

An observation is defined as a unique triple consisting of a disrupted firm i (or a customer/supplier in Panel 2 and 3), a disruption announcement date d, and a trading day t relative to the announcement date. Firms in our data show a notable presence of multisourcing and customer diversification, and a median relationship length exceeds 1 year. Measures MS and CD are right-skewed, thus we use $\ln(MS+1)$ and $\ln(CD+1)$ in regressions. The distributions of variables show a considerable heterogeneity across industries. The histograms for two representative industries, Oil And Gas Extraction (SIC2: 13) and Chemicals And Allied Products (SIC2: 28), demonstrate heterogeneity within industries and differences in shapes of the distributions. The samples in these two industries account for around 10% of total sample for disrupted firms, customers, and suppliers. The values of the histogram's x-axis range from 0 to 5. Because of the heterogeneity at the level of firms, we use firm-level fixed effects in our models. We also report summary statistics for the log-transformed (to address pronounced right skewness) market-to-book ratio (MB), $MCAP$, LIQ, and VLT when available. For presentation purposes, their histograms are displayed using six frequency-based bins rather than a fixed 0-5 range.

7. Results

We start with presenting the regression analysis of post-disruption returns and their interaction with sourcing strategy variables. We proceed with the analysis of marginal effects of sourcing strategies on firms' value and formal hypotheses testing. The section concludes with the analysis of the subsample of disasters which mitigates endogeneity concerns in estimating the effect of sourcing strategy on value resilience.

Figure 6 **The value destruction and the time to recover market value as a function of buyer-supplier relationship length, for focal firms (a), customers (b), and suppliers (c)**



7.1. Disrupted Firms

Table 7 presents the regression results for equation (3) and (4). We observe a U-shaped relationship between time elapsed after a disruption and the firm's stock return, further moderated by the average relationship length $\overline{RL}$, increasing the linear in t term.

To visualize how returns as well as $TTRMV/MVD$ change with different relationship lengths (Figure 6(a)), we plug the 90th, 75th, 50th, and 25th percentiles of $\overline{RL}$ into the fitted model (4) using column (2). We also include "Fitted" line where variables (i.e, $\overline{RL}$ and controls) are set at their mean values. Figure 6(a) illustrates that disrupted firms with longer supply chain relationships generally recover quicker, as indicated by a shorter TTRMV. Interestingly, all fitted lines intersect at some t', which is around 20 days. That is because the relationship length $\overline{RL}$ enters (4) multiplicatively with t as $\overline{RL}(\beta_3 + \beta_4 t + \beta_5 t^2)$. The function in parenthesis is quadratic, hence there exist $t'_{1,2}$, such that $\beta_3 + \beta_4 t' + \beta_5 t'^2 = 0$, and all lines intersect. At those points, the expected cumulative returns are independent of $\overline{RL}$, yet the trajectories diverge in the future. This suggests an arbitrage opportunity, going long in a firm with higher $\overline{RL}$, and shorting one with lower $\overline{RL}$. However, to execute the strategy multiple firms need to be disrupted at the same time. We discuss additional arbitrage opportunities in §8.

7.2. Customers and Suppliers of Disrupted Firms

Table 8, columns (1)-(3) presents the analysis of value recovery dynamics of customers for specifications (5) and (6), the latter with and without focal returns. $focal\_BAHCR$ is highly significant, supporting downstream return comovement in Hypothesis 1. Table 8 also shows that relationship length is significantly associated with the returns of the customers following disruptions. To visualize how returns, MVD, and $TTRMV$ change with relationship lengths for customers, we follow the same approach as for the disrupted firms using (6) without focal firm returns (i.e., column (3) of Table 8). Figure 6(b) suggests that customers with longer relationships with disrupted firms recover faster from propagated disruptions and experience market value losses of smaller magnitude.

Table 8 Association between sourcing strategies and post disruption returns for customers and suppliers.

	Customers				*Suppliers*		
	(1) BAHCR (5)	(2) BAHCR (6)	(3) BAHCR (6)		(4) BAHCR (5)	(5) BAHCR (6)	(6) BAHCR (6)
t	-4.10	-8.34	-10.35	t	-3.13	-9.17	-11.78
	(6.60)	(7.85)	(8.90)		(4.75)	(6.15)	(8.17)
t^2	0.08	0.16^{*}	0.19^{**}	t^2	0.07	0.13^{***}	0.18^{***}
	(0.06)	(0.07)	(0.06)		(0.04)	(0.03)	(0.05)
RL	1.62	-3.55	-4.50	RL	14.23	-13.71	-13.15
	(9.62)	(13.50)	(15.75)		(9.73)	(15.58)	(18.01)
$\ln(MS+1)$	34.17	12.89	17.36	$\ln(CD+1)$	-10.77	-53.77	-51.97
	(31.31)	(39.33)	(38.76)		(36.94)	(37.49)	(41.94)
$t \times RL$		1.39^{*}	1.81^{+}	$t \times RL$		1.98^{**}	2.14^{*}
		(0.66)	(0.92)			(0.69)	(0.84)
$t \times \ln(MS+1)$		1.27	1.40	$t \times \ln(CD+1)$		1.35	2.08
		(1.70)	(1.51)			(2.06)	(2.50)
$t^2 \times RL$		-0.03^{**}	-0.03^{*}	$t^2 \times RL$		-0.03^{*}	-0.03^{*}
		(0.01)	(0.02)			(0.01)	(0.01)
$t^2 \times \ln(MS+1)$		-0.01	-0.01	$t^2 \times \ln(CD+1)$		0.00	-0.01
		(0.03)	(0.02)			(0.03)	(0.04)
$focal\_BAHCR$	0.25^{***}	0.25^{***}		$focal\_BAHCR$	0.23^{***}	0.23^{***}	
	(0.04)	(0.04)			(0.06)	(0.06)	
N	270787	270787	270787	N	509882	509882	509882
Customer FE	1433	1433	1433	Supplier FE	2239	2239	2239
Year-Quarter FE	66	66	66	Year-Quarter FE	66	66	66
Controls	✓	✓	✓	Controls	✓	✓	✓
R^2	0.42	0.42	0.39	R^2	0.37	0.37	0.34

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$.

The value recovery dynamics for suppliers is presented in columns (4)-(6) of Table 8. The significant coefficients of $focal\_BAHCR$ provide support for upstream returns comovement; together with downstream comovement, these findings present full evidence for Hypothesis 1. Table 8 also reveals significant association between relationship length and supplier returns post-disruption. Figure 6(c) shows how returns, MVD, and $TTRMV$ evolve with relationship length for suppliers, based on equation (6) (column (6) in Table 8). The figure indicates that longer supplier relationships are associated with faster recovery and smaller losses. Similarly to the disrupted firms, at time t' (between 5–10 days), $\beta_3 + \beta_5 t' + \beta_7 t'^2 = 0$, and all curves intersect.

Investigating the effect of SCAD, we find it to be insignificant. Next, we quantify the association between sourcing strategies and $TTRMV/MVD$ among disrupted firms, their customers and suppliers in §7.3 and test Hypotheses 2-4.

7.3. Effectiveness of Risk Mitigation Strategies

Because MVD and $TTRMV$ are functions of the estimated regression coefficients, we use bootstrapping to estimate the sensitivity of MVD and $TTRMV$ to changes in variables of interest (i.e., $\overline{RL}$, RL, MS, and CD), along with standard errors of these sensitivities. We repeat our estimates on 1000 iterations with samples drawn with replacement from the full population for disrupted firms/customers/suppliers. For disrupted firms, the resampling is conducted at $i-d$ level (i.e., a pair of a disrupted firm and an announcement date); for customers/suppliers, the resampling is conducted at $i-d-focal$ level (i.e., a triple of a customer/supplier, announcement date, and disrupted firm). In each iteration, we estimate the following sensitivities of $TTRMV/MVD$ to variables of interests:

$$\frac{\partial MVD}{\partial X} = \frac{\partial\{A\cdot T^2 + B\cdot T + C\}}{\partial X}; \; \frac{\partial TTRMV}{\partial X} = \frac{\partial\{\frac{-B+\sqrt{B^2-4AC}}{2A}\}}{\partial X}, \tag{7}$$

where A, B, and C are calibrated according to bootstrapping pool samples (disrupted firms/customers/suppliers). In iterations for disrupted firms, $A = \beta_2 + \beta_5\overline{RL}$, $B = \beta_1 + \beta_4\overline{RL}$, $C = \alpha + \delta + \beta_3\overline{RL} + \beta\mathbf{M}$, and X is $\overline{RL}$. In iterations for customers and suppliers, $A = \beta_2 + \beta_7 RL + \beta_8 \ln(SDS+1)$, $B = \beta_1 + \beta_5 RL + \beta_6 \ln(SDS+1)$, and $C = \alpha + \delta + \beta_3 RL + \beta_4 \ln(SDS+1) + \beta\mathbf{M}$. We use $SDS = MS$ for customers with X from $\{MS, RL\}$ and $SDS = CD$ for suppliers with X from $\{CD, RL\}$. The rest of the parameters are the same as in model (4) and (6), the latter without focal returns. Since our fitted models are U-shaped, we can solve $At^2 + Bt + C$ to get $TTRMV/MVD$ and $T = -\frac{B}{2A}$ accordingly. In (7), variables not conditioned on (i.e., non X) are set to the sample average in each bootstrapped sample.

Table 9 The marginal effects of sourcing strategies (long-term relationships, multisourcing, and customer diversification) on market value resilience ($TTRMV$ and MVD).

Panel 1: Disrupted Firms (Resampled at Firm-Date Level)

Resampling Pool Size	753	
	Mean Est.	St.Err.
$\partial MVD/\partial\overline{RL}$, bps/year	13.01	74.51
$\partial TTRMV/\partial\overline{RL}$, days/year	-15.63**	6.03

Panel 2: Customers (Resampled at Customer-Date Level)

Resampling Pool Size	4781	
$\partial MVD/\partial RL$, bps/year	18.90***	3.54
$\partial TTRMV/\partial RL$, days/year	-0.73	1.60
$\partial MVD/\partial MS$, bps/supplier	25.32***	6.28
$\partial TTRMV/\partial MS$, days/supplier	-7.34***	1.84

Panel 3: Suppliers (Resampled at Supplier-Focal-Date Level)

Resampling Pool Size	9196	
$\partial MVD/\partial RL$, bps/year	21.30***	3.66
$\partial TTRMV/\partial RL$, days/year	-6.21***	1.44
$\partial MVD/\partial CD$, bps/customer	-3.29	5.02
$\partial TTRMV/\partial CD$, days/customer	-2.80*	1.38

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$.

Table 9 details the bootstrapping results. For disrupted firms, each additional year of RL is associated with about 15.6 fewer days to recover market value. For customers, each additional year of RL with a focal firm (experiencing a disruption) is associated with an 18.90 bps increase in MVD (a higher value indicates less absolute value destruction). For suppliers, each additional year of RL is negatively associated with $TTRMV$ (-6.21 days) and positively associated with MVD (21.30

bps). Therefore, our findings support Hypothesis 2 for disrupted firms as well as their customers and suppliers.

Sourcing strategies also show distinct associations. For suppliers, a higher number of alternative customers (CD) is linked to a shorter $TTRMV$ (-2.80 days per customer), partially supporting Hypothesis 4. For customers, the presence of an additional alternative supplier in the same industry as the disrupted firm correlates with a 25.32 bps higher MVD and a 7.34-day reduction in $TTRMV$, supporting Hypothesis 3.

These findings suggest that long-term relationships correlate with more favorable outcomes for disrupted firms, as shown in significantly shorter $TTRMV$. This suggests that longer-standing ties may help disrupted firms stabilize operations and access critical resources during disruptions, facilitating faster value recovery. Likewise, for customers, longer RL correlates with smaller value losses. This pattern may reflect strategic efforts by disrupted firms (acting as suppliers) to stabilize sales channels. To preserve critical downstream revenue, these firms may prioritize coordination with long-term customers, mitigating the market value loss. Similarly, for the suppliers of disrupted firms, the negative association between RL and $TTRMV$ and the positive association with MVD suggest that longer ties are linked to both shielded firm value and faster recovery. This is consistent with the interpretation that suppliers with established ties are more attentive to the stability of their customers and may coordinate more closely with disrupted partners to restore operational continuity.

The results also highlight the associations between sourcing strategies and market value resilience. For customers, multisourcing is linked to faster recovery and reduced value loss, consistent with the benefits of redundant supply options during a partner's disruption (supporting Hypothesis 3). For suppliers, the observation that $TTRMV$ decreases as the customer base diversifies suggests that a broader downstream network may help spread demand risk. By maintaining multiple relationships, suppliers may reduce their dependence on any single disrupted firm. Notably, the lack of a significant association between diversification and MVD suggests that while a diverse customer base is associated with faster recovery, it may not necessarily relate to the magnitude of the initial shock. This provides nuanced support for Hypothesis 4.

7.4. Analysis of Disaster Events

Disruptions may manifest in different forms. Building on the keywords listed in Table 1, we flag disruption announcements based on whether they mention keywords associated with 4 broad dimensions: supply-, logistics-, quality-, and natural disaster-related issues. This process reveals noticeable heterogeneity in recovery times with longer $TTRMV$ in supply, logistics, and disaster-related disruptions (Table 5).

Table 10 Marginal effects of sourcing strategies for the disaster-related subsample.

Panel 1: Disrupted Firms (Resampled at Firm-Date Level)

	Mean Est.	St.Err.
Resampling Pool Size	394	
$\partial MVD/\partial \overline{RL}$, bps/year	557.51***	106.98
$\partial TTRMV/\partial \overline{RL}$, days/year	7.42	33.26

Panel 2: Customers (Resampled at Customer-Focal-Date Level)

	Mean Est.	St.Err.
Resampling Pool Size	2580	
$\partial MVD/\partial RL$, bps/year	20.39***	5.95
$\partial TTRMV/\partial RL$, days/year	5.11***	1.19
$\partial MVD/\partial MS$, bps/supplier	27.64***	8.36
$\partial TTRMV/\partial MS$, days/supplier	2.88$^+$	1.50

Panel 3: Suppliers (Resampled at Supplier-Focal-Date Level)

	Mean Est.	St.Err.
Resampling Pool Size	4196	
$\partial MVD/\partial RL$, bps/year	21.30**	6.83
$\partial TTRMV/\partial RL$, days/year	1.33	1.41
$\partial MVD/\partial CD$, bps/customer	8.76	7.53
$\partial TTRMV/\partial CD$, days/customer	2.64*	1.32

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$.

We focus on the natural or technological disasters. The disasters are large scale unanticipated events that affect firms in geographical proximity. Because the timing and intensity of disasters are plausibly exogenous conditional on firm fixed effects, this design mitigates concerns that unobserved firm capabilities jointly determine sourcing strategies, disruption exposure, and stock performance.

For disrupted firms and their customers/suppliers, we estimate regressions (4) and (6) (the latter without focal firm returns), respectively, with the same controls in each subsample (see Table EC.3). The estimated coefficients are broadly consistent with those reported in Tables 7 and 8.

We report the marginal effects of sourcing strategies on the sample of disaster events in Table 10. We observe a stronger marginal effect of $\overline{RL}$ on MVD for disrupted firms: having one year longer relationship is associated with 557.71 bps smaller value loss, compared to 13.01 bps (insignificant) in the general sample. The marginal effects for suppliers and customers are similar and comparable to the general sample. The marginal effects on $TTRMV$ are weaker but directionally positive, suggesting a trade-off between milder MVD and longer $TTRMV$ on the disaster subsample.

8. Implications for Firms and Investors

The results of §7 show that $TTRMV$ and MVD, the measures of market value resilience, respond to changes in supply network design. In this section, we investigate whether they also associate with changes in the cost of debt. We also provide causal evidence that relationship length affects the post disruption cost of debt. We conclude with a discussion of regularities in short term returns and long-term valuations.

8.1. Cost of Debt

We examine whether resilience to disruptions is associated with more favorable financing conditions, specifically a lower cost of debt, and whether a longer RL, a precursor to resilience, is causally related to a lower cost of debt. For disrupted firm i in year-quarter k, we estimate a difference-in-differences specification (8) around disruption events. Firms are assigned to the treatment and control groups as follows. The sample is split into terciles based on $TTRMV$, MVD, or $\overline{RL}$ at the time of disruption. For example, $Long\_RL = 1$ would indicate the top tercile and $Long\_RL = 0$ for the bottom tercile (middle tercile is dropped). Firm FE α_i controls for time-invariant unobservable firm characteristics. Year-quarter FE of disruption announcement date δ_{yq} controls for the market performance in that year-quarter and other common shocks.

$$r_{Dik} = \alpha_i + \delta_{yq} + \beta_1 Post_{ik} + \beta_2 Treated_{ik} + \beta_3 Treated_{ik} \times Post_{ik} + \epsilon_{i,yq}. \tag{8}$$

The results are presented in Table 11. The first result for $TTRMV$ shows a significant association between fast recovery of market value and a lower cost of debt (measured as an average interest rate over outstanding debt) post disruption. The second result for MVD shows that a milder value destruction is insignificant, yet the direction is consistent with the result on $TTRMV$. An important caveat to these results is that $TTRMV$ and MVD are not causal factors but, instead, are realized market reactions.

To demonstrate a causal relationship to the cost of debt, we use the assignment to treatment based on $\overline{RL}$, one of the supply network design variables associated with the market value resilience. The results show that longer buyer-supplier relationships result in a lower cost of debt in post disruption quarters (Table 11, Panel C). We verify the parallel trends assumption and conduct falsification tests showing that the results significantly differ from the ones where assignment to treatment is at random, supporting the validity of the difference-in-differences specification, see E-Companion EC.8.

8.2. Short-term Returns

Table 9 shows that, following a disruption, customers/suppliers with longer relationships with disrupted firms experience systematically higher stock returns. Our analysis of cumulative raw returns corroborates this pattern and reveals substantial heterogeneity across customers based on relationship length. Specifically, 20 trading days after a disruption announcement, customers in the bottom quartile of relationship length (≤ 0.67 years) exhibit an average return of -5.62%, whereas those in the top quartile (≥ 3.13 years) exhibit an average return of 1.63%, yielding a statistically significant spread of 7.25% ($p < 0.05$). This pronounced cross-sectional return differential suggests that relationship duration meaningfully predicts how customer firms are revalued in the short run

Table 11 Implications of long-term buyer-supplier relationships for the cost of debt.

Panel A		Panel B		Panel C	
	$\ln r_D$		$\ln r_D$		$\ln r_D$
Post	0.001	*Post*	−0.010	*Post*	0.016
	(0.025)		(0.028)		(0.023)
Fast_TTRMV	0.003	*Mild_MVD*	−0.002	*Long_RL*	0.053
	(0.037)		(0.036)		(0.054)
Fast_TTRMV × *Post*	-0.061^{+}	*Mild_MVD* × *Post*	−0.037	*Long_RL* × *Post*	-0.084^{*}
	(0.036)		(0.036)		(0.038)
N	3207	N	3207	N	3836
Firm FE	272	Firm FE	272	Firm FE	287
Year-Quarter FE	65	Year-Quarter FE	65	Year-Quarter FE	65
R^2	0.746	R^2	0.745	R^2	0.788

$^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$

following supply chain disruptions. Importantly, such heterogeneity is not uncommon: 175 out of the 852 disruption events in our sample exhibit sufficiently wide dispersion in customer relationship length to generate this effect. The return spread emerges over a relatively short horizon, highlighting the speed at which relationship-specific information is gradually reflected in prices.

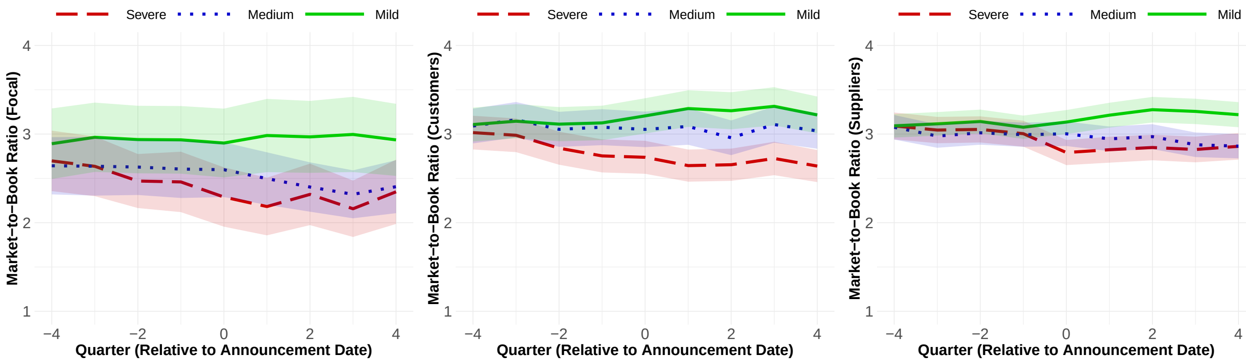
8.3. Long-term Valuations

We examine longer-horizon valuation outcomes by comparing firm performance before and after disruptions, focusing on differences across firms with faster versus slower recovery and milder versus more severe value destruction. Specifically, we compute firms' Market-to-Book (MB) ratios in the four quarters before and after disruption events. The MB ratio captures the market's valuation of a firm relative to its book value and is commonly interpreted as reflecting expectations about future profitability, growth opportunities, or intangible assets.

Figure 7 plots average MB ratios for firms grouped into terciles based on MVD (BAHCR-based) following supply chain disruptions. Firms in the severe, medium, and mild terciles correspond to those experiencing the largest (red dashed lines), intermediate (blue dotted lines), and smallest (green solid lines) MVD magnitude, respectively. Across all panels, firms in the mild MVD tercile exhibit persistently higher post-disruption MB ratios, with the pattern most pronounced among demand-disrupted suppliers. At the same time, firms in the severe MVD tercile show drops in the MB ratios, lasting for four quarters or longer, a period of time exceeding the average $TTRMV$ in that tercile.

For focal firms, we further observe that those experiencing milder MVD already display higher MB ratios even prior to the disruption announcement quarter. This pre-trend is consistent with Hendricks and Singhal (2005), which documents that the underperformance of disrupted firms can be reflected in financial information well before disruption announcement. More broadly, the post-disruption dynamics extend beyond focal firms: focal firms, customers, and suppliers with milder value losses all exhibit persistently and significantly higher MB ratios after the disruption event.

Figure 7 **Market-to-book ratio of focal firms, their customers and suppliers before and after disruption, by disruption severity.**



Importantly, the divergence in MB ratios persists for up to four quarters afterward, a much greater period than the 60-day window for MVD computation.

We further examine these patterns using regressions. Table EC.4 reports estimates relating post-disruption MB ratios to disruption severity and recovery speed. For disrupted focal firms, customers, and suppliers, a 1% (100 bps) reduction in maximum value loss (i.e., a 1% increase in MVD) is associated with a 0.7%, 0.6%, and 0.5% higher post-disruption MB ratio, respectively. The trend aligns with the descriptive evidence in Figure 7 and suggests that differences in market value resilience to disruptions may be reflected in longer-term market valuations.

9. Conclusion

In this paper, we examine the role of long-term buyer-supplier relationships, multisourcing and customer diversification in the resilience of firms' market value in the aftermath of supply chain disruptions. We propose two measures of the impact of disruptions on the market value: maximum value destruction (MVD) and time to recovery of market value ($TTRMV$). We document that disruption events are associated with significant market value losses for disrupted firms as well as their suppliers and customers that take several weeks to recover.

In an event study framework, we show that the valuation impact of disruptions varies systematically with sourcing strategies. Longer buyer-supplier relationships are associated with milder value losses and faster recovery, while multisourcing and customer diversification predict higher market value resilience for suppliers and customers. We further validate these findings on a subsample of disaster-related disruptions, showing a stronger relationship between the relationship length and maximum value destruction, and mitigating the endogeneity concerns between relationship length and disruption occurrence. Among disrupted firms, a longer buyer-supplier relationships are associated with greater market value resilience and lead to a lower cost of debt in post-disruption periods. Finally, firms with high resilience to disruptions, their customers, and suppliers exhibit higher market-to-book ratios in post-disruption quarters.

While conducting this study, we develop a fine-tuned LLM-based workflow for identifying supply chain disruptions from firm press releases. The fine-tuned model performs comparably to general-purpose LLMs (e.g., GPT-4o) on this task, illustrating the promise of custom-built AI models for improving resilience of supply chains.

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E-Companion to "Value Destruction and Recovery from Supply Chain Disruptions"

EC.1. Details of Fine-Tuning

EC.1.1. Fine-Tuning

For fine-tuning, there are two options: starting from a pretrained model or an instruct model. Pretrained models are trained on large-scale text corpus using self-supervised learning to predict the next token in a sequence. Instruct models build on these pretrained models and are further fine-tuned to follow human instructions in a conversational or task-oriented manner. We fine-tuned a Llama 3 8B *pretrained model*, instead of an instruct model, because of the following reasons. First, we aim to build a text classifier that directly maps raw articles to binary labels without relying on prompt-based inputs. This design ensures that the classifier produces consistent outputs and is not sensitive to prompt variations or phrasing choices that often affect instruct models. Second, instruct models rely on conversational prompts and multi-domain labeler knowledge during training, introducing unnecessary variability for our task. Their generalized alignment process is suboptimal for the identification of supply chain disruption events. Moreover, instruction-following behavior of instruct models may lead to hallucination.

We applied low-rank adaptation (Hu et al. 2022), a parameter-efficient fine-tuning method, to fine-tune an 8-bit quantized Llama 3 8B pretrained model. Low-precision (i.e., 8-bit) quantization reduces memory for fine-tuning. We used 70% of the manual tagging sample for fine-tuning model training process, and 15% for model validation. The remaining 15% texts unseen by the model are used for test. The fine-tuning is based on AUC (Area Under Receiver Operating Characteristic Curve) score. Using an NVIDIA A100 80GB GPU in a HPC cluster, we trained the model for 30 epochs over 16.85 hours. The best-performing checkpoint achieved a substantial improvement in AUC, increasing from 53.70% (original pretrained model) to 96.25% (fine-tuned model), as shown in Figure 1. Using the argmax of the logits, which is equivalent to applying a 0.5 cutoff on the softmax output in a binary classification setting, we construct the confusion matrix as shown in Table 2. There are 43 texts labeled as disruptions (Class 1) and 145 texts labeled as non-disruptions (Class 0). Precision and recall can be calculated accordingly. For Class 1, the model achieved a precision of 89% and a recall of 77%. For Class 0, precision was 93% and recall was 97%.

EC.1.2. Performance of Fine-Tuning

Ideally, we aim to compare model performance on cases where predictions disagree, i.e., instances where the fine-tuned model (After), the generic model (Before), and GPT-4o yield different classifications. However, our test dataset contains only 24 disagreements between GPT-4o and After, and 56 between Before and After, insufficient for reliable bootstrapping. Instead, we bootstrap the entire test set (181 samples) with replacement over 1000 iterations. In each iteration, we compute each model's precision, recall, and Cohen's kappa (excluding AUC, as reliable prediction probabilities for GPT-4o are not publicly accessible), and record whether After yields higher values than Before or GPT-4o on each evaluated metric (1 = yes, 0 = no). Across all iterations, we derive the mean, standard error, 95% confidence interval (CI), and p-value for pairwise comparisons. Table EC.1 shows that fine-tuning substantially improves performance over the generic model and achieves results comparable to commercial models on specialized tasks.

Table EC.1 Paired Performance of Models

Comparison	Metric	Mean	SE	95% CI	p value
After > Before	Precision	1.000	0.000	1.000 - 1.000	0.000
After > Before	Recall	1.000	0.000	1.000 - 1.000	0.000
After > Before	Cohen's kappa	1.000	0.000	1.000 - 1.000	0.000
After > GPT-4o	Precision	0.860	0.011	0.838 - 0.882	0.000
After > GPT-4o	Recall	0.563	0.016	0.532 - 0.594	0.000
After > GPT-4o	Cohen's kappa	0.767	0.013	0.741 - 0.793	0.000

EC.2. Firm Coverage in FactSet Data

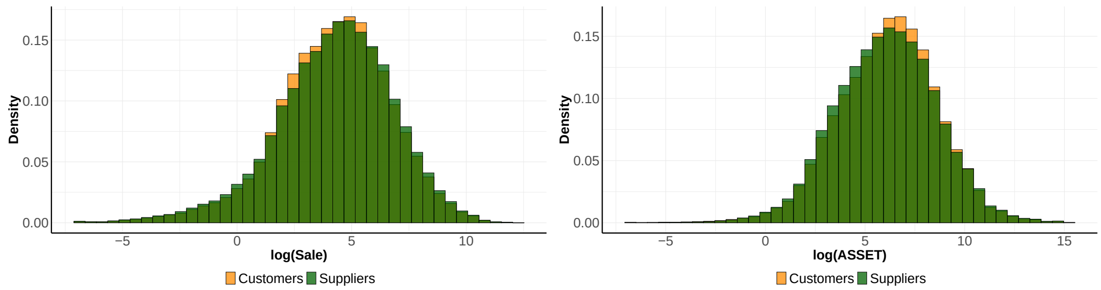
To assess whether asymmetric coverage in FactSet could mechanically drive the observed downstream-upstream asymmetry, we conducted a systematic validation of firm size coverage across supply-chain roles. Motivated by the concern that FactSet relationships may disproportionately capture larger downstream partners, we assembled the full universe of FactSet-identified buyer-supplier relationships over our sample period (2003-04-03 to 2019-12-31) and linked all firms to Compustat Fundamentals Quarterly. This procedure yields 10461 unique CUSIPs with both FactSet and Compustat information, of which 9327 have ever appeared as customers and 8297 as suppliers. Using total sales and total assets as size proxies, we compare the log-transformed distributions for customers and suppliers. Figure EC.1 exhibits substantial overlap for both measures, indicating that firms identified as customers are not systematically larger than those identified as suppliers, alleviating concerns that selection on firm size is driving the documented asymmetry in disruption spillovers.

EC.3. Matching Disruptions to Network and Stock Return Data

EC.3.1. Ticker Matching

In many Factiva press releases, public firm names are formatted as follows: "Firm Name [Exchange Code: Ticker]". For example, from the first entry in Table 3 we can directly extract the ticker of Georgia Gulf Corporation by locating "GGC" in "[NYSE: GGC]". Because tickers can change over time we also use the press release year for matching. This approach yields 597 articles, or 663

Figure EC.1 Customers' versus Suppliers' Size in FactSet Revere Relationship Data (2003-04-03 to 2019-12-31)



article-ticker pairs, since multiple public firms can be mentioned in some texts. If a press release is classified as disruption, we assume that all firms mentioned there are disrupted. This approach is conservative, because a potential inclusion of non-disrupted firms would dilute our results.

EC.3.2. Firm Name Matching

If tickers are not available, we use the firm name (reported by Factiva) and match it to the $COMNAM$ data field in CRSP. We consider the following variants for matching: (i) the original COMNAM, (ii) adding a comma (e.g., "Apple Inc"→"Apple, Inc"), (iii) expanding suffixes such as "ltd" to "limited", (iv) removing spaces around symbol "&" (e.g., "W & T OFFSHORE INC"→"W&T OFFSHORE INC") for firm names with single letters. This approach yields 1,126 texts, which partially overlap with the ticker matched set. Because we match the full texts of press releases, we do not strip incorporating suffixes, as doing so could lead to false positive matches, e.g., Apple, Inc. could be matched with "apple" and Target Corp. with "target". A combination of the two approaches in total gives 1,197 articles, or 1,410 article-CUSIP pairs. After removing articles describing the same disruption, we are left with 1,166 articles, or 1,368 article-CUSIP pairs.

EC.4. Evidence of Disruption Impact using Abnormal Returns

Figures EC.2, EC.3 and EC.4 present cumulative abnormal returns of disrupted firms, customers and suppliers following disruption announcements.

Figure EC.2 Cumulative Abnormal Returns of Disrupted Firms in Basis Points

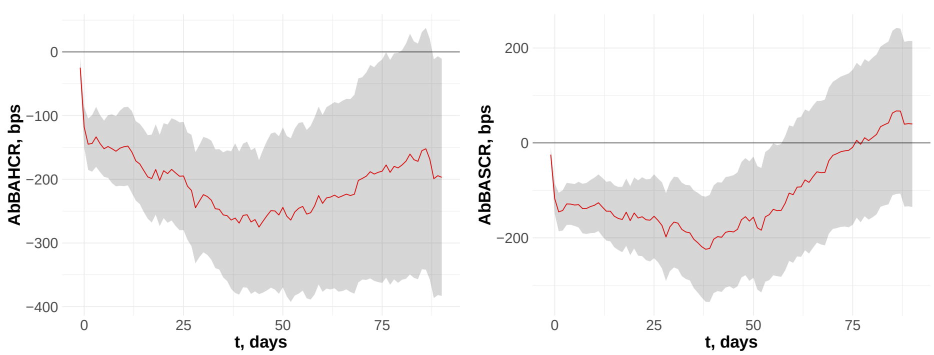


Figure EC.3 Cumulative Abnormal Returns of Customers in Basis Points

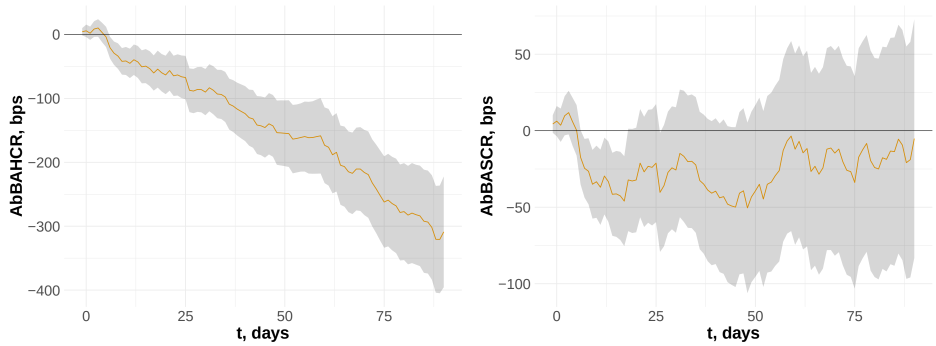
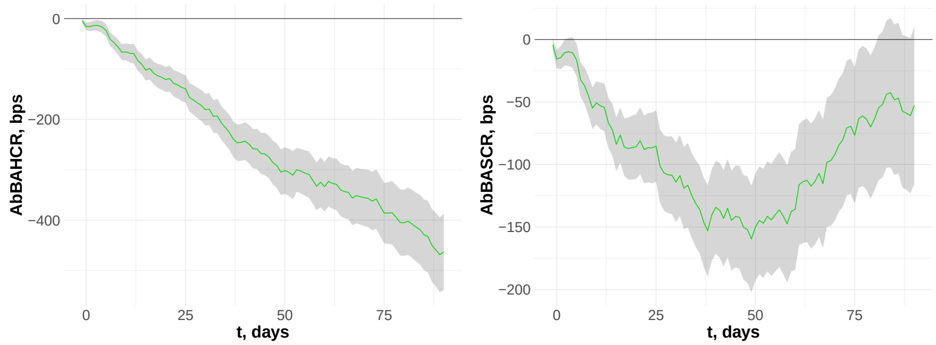


Figure EC.4 Cumulative Abnormal Returns of Suppliers in Basis Points



EC.5. Connection to Short-Term Returns

Following Hendricks and Singhal (2003), we regress firms' time to recover market value ($TTRMV$) and maximum value destruction (MVD) on cumulative (abnormal) returns measured over a 3-day window surrounding the disruption announcement, as shown in Table EC.2.

EC.6. Heterogeneity of Disruptions: Subsample Regressions

In each subsample regression, we only include firms (disrupted or customers/suppliers) that are involved in specific disruption announcements. Table EC.3 presents the subsample regression results.

Table EC.2 Predicting TTRMV and MVD Using 3-Day Window Returns

Panel 1: Disrupted Firms

	$TTRMV$ (1)	$TTRMV$ (2)	$TTRMV$ (3)	$TTRMV$ (4)	MVD (1)	MVD (2)	MVD (3)	MVD (4)
(Intercept)	12.27***	12.10***	11.98***	11.66***	−0.05***	−0.06***	−0.04***	−0.05***
	(0.44)	(0.43)	(0.39)	(0.39)	(0.00)	(0.00)	(0.00)	(0.00)
$BAHCR$	−35.15***				0.64***			
	(8.70)				(0.09)			
$BASCR$		−37.19***				1.00***		
		(8.45)				(0.19)		
$AbBAHCR$			−50.66***				0.77***	
			(9.36)				(0.11)	
$AbBASCR$				−42.96***				0.98***
				(8.59)				(0.19)
N	999	1059	979	1008	999	1059	979	1008
R^2	0.02	0.03	0.05	0.04	0.26	0.28	0.34	0.32
Panel 2: Customers								
(Intercept)	11.33***	11.24***	10.92***	10.95***	−0.04***	−0.05***	−0.03***	−0.04***
	(0.27)	(0.27)	(0.21)	(0.23)	(0.00)	(0.00)	(0.00)	(0.00)
$BAHCR$	−13.11+				0.53***			
	(7.04)				(0.07)			
$BASCR$		−18.59**				0.73***		
		(6.84)				(0.09)		
$AbBAHCR$			−18.47**				0.56***	
			(7.15)				(0.05)	
$AbBASCR$				−29.11***				0.76***
				(7.48)				(0.09)
N	4351	4464	4137	4210	4351	4464	4137	4210
R^2	0.00	0.00	0.00	0.01	0.13	0.14	0.15	0.13
Panel 3: Suppliers								
(Intercept)	10.99***	11.08***	10.33***	10.39***	−0.05***	−0.05***	−0.04***	−0.04***
	(0.22)	(0.25)	(0.18)	(0.18)	(0.00)	(0.00)	(0.00)	(0.00)
$BAHCR$	−13.22+				0.52***			
	(7.45)				(0.09)			
$BASCR$		−19.97**				0.68***		
		(7.63)				(0.11)		
$AbBAHCR$			−27.84***				0.57***	
			(5.62)				(0.04)	
$AbBASCR$				−31.01***				0.64***
				(5.15)				(0.06)
N	8190	8438	7716	7908	8190	8438	7716	7908
R^2	0.00	0.01	0.01	0.01	0.15	0.16	0.18	0.14

Note. $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$. Regression coefficients are based on raw returns, rather than basis points (i.e., a return of 0.1 represents 10%). The numbers in column parenthesis: 1/2/3/4, metrics calculated based on BAHCR/BASCR/AbBAHCR/AbBASCR respectively.

Table EC.3 Associations Between Supply Chain Relationships and Recovery of Firms

	BAHCR (Supply)	BAHCR (Logistics)	BAHCR (Quality)	BAHCR (Disaster)
Panel 1: Focal Disrupted Firms				
t	-42.51^{**}	-32.23^{*}	-14.42	-31.37^{+}
	(15.82)	(13.98)	(19.07)	(16.51)
t^2	0.69^{**}	0.52^{**}	-0.21	0.49^{**}
	(0.23)	(0.16)	(0.53)	(0.16)
$\overline{RL}$	483.35	-307.84	6243.93	393.86
	(840.38)	(327.09)	(4983.28)	(360.55)
$t\times\overline{RL}$	18.27^{**}	13.25^{+}	12.20	13.23
	(6.67)	(7.46)	(8.98)	(8.78)
$t^2\times\overline{RL}$	-0.26^{*}	-0.23^{*}	0.24	-0.22^{+}
	(0.11)	(0.10)	(0.38)	(0.12)
N	15311	24766	10065	22509
Disrupted Firm FE	174	246	89	212
Year-Quarter FE	61	65	47	62
Controls	✓	✓	✓	✓
R^2	0.65	0.63	0.48	0.64
Panel 2: Customers of Disrupted Firms				
	BAHCR (Supply)	BAHCR (Logistics)	BAHCR (Quality)	BAHCR (Disaster)
t	-9.44	-17.97	8.51^{+}	-21.87
	(6.46)	(11.71)	(5.08)	(14.11)
t^2	0.20^{*}	0.23^{*}	0.03	0.31^{**}
	(0.09)	(0.09)	(0.09)	(0.09)
RL	-16.10	-18.82	0.39	-10.83
	(16.92)	(17.61)	(26.27)	(18.97)
$\ln(MS+1)$	77.72	-97.37^{+}	-176.99	-12.30
	(65.12)	(48.83)	(125.94)	(48.64)
$t\times RL$	2.10^{*}	3.21^{***}	0.22	2.40^{+}
	(0.81)	(0.82)	(1.24)	(1.26)
$t\times\ln(MS+1)$	2.53	5.13^{*}	5.66	4.13^{+}
	(2.93)	(2.09)	(3.65)	(2.13)
$t^2\times RL$	-0.05^{**}	-0.06^{***}	-0.02	-0.04^{*}
	(0.02)	(0.01)	(0.02)	(0.02)
$t^2\times\ln(MS+1)$	-0.03	-0.01	-0.09^{*}	-0.06^{+}
	(0.06)	(0.03)	(0.05)	(0.03)
N	106002	153875	52572	145493
Customer FE	878	1044	472	1013
Year-Quarter FE	62	64	52	64
Controls	✓	✓	✓	✓
R^2	0.46	0.49	0.47	0.52
Panel 3: Suppliers of Disrupted Firms				
	BAHCR (Supply)	BAHCR (Logistics)	BAHCR (Quality)	BAHCR (Disaster)
t	-16.67^{+}	-14.06	6.66	-24.85
	(8.80)	(10.30)	(4.52)	(16.48)
t^2	0.28^{**}	0.22^{**}	-0.00	0.29^{**}
	(0.09)	(0.07)	(0.08)	(0.10)
RL	-18.29	-32.33	26.75	-31.62
	(20.39)	(20.93)	(25.85)	(33.38)
$\ln(CD+1)$	16.13	-69.24	-32.83	-149.44^{*}
	(70.57)	(65.38)	(59.88)	(74.64)
$t\times RL$	3.10^{**}	3.22^{**}	0.79	2.37
	(0.93)	(0.99)	(0.67)	(1.50)
$t\times\ln(CD+1)$	3.84	3.12	0.83	7.39^{*}
	(5.17)	(3.86)	(2.67)	(3.08)
$t^2\times RL$	-0.06^{*}	-0.05^{*}	0.00	-0.03^{+}
	(0.02)	(0.02)	(0.01)	(0.01)
$t^2\times\ln(CD+1)$	0.02	-0.03	-0.02	-0.08^{*}
	(0.07)	(0.05)	(0.04)	(0.04)
N	169433	262228	172299	231458
Supplier FE	1367	1725	1092	1464
Year-Quarter FE	62	66	50	64
Controls	✓	✓	✓	✓
R^2	0.47	0.44	0.31	0.49

$^{***}p<0.001$; $^{**}p<0.01$; $^{*}p<0.05$; $^{+}p<0.1$

EC.7. Additional Market-to-Book Analyses

We use regression (EC.1) to examine the association between MVD and post-disruption MB ratio. Firm FE is controlled by α_i. q is the year-quarter of calculated MB.

$$\ln MB_{iq} = \alpha_i + \beta_1 Post_{iq} + \beta_2 MVD_{iq} + \beta_3 MVD_{iq} \times Post_{iq} + \epsilon_{iq} \tag{EC.1}$$

Table EC.4 Use MVD to Predict MB

	$\ln MB$ (Focal)	$\ln MB$ (Customers)	$\ln MB$ (Suppliers)
$Post$	-0.00831	0.00239	0.00030
	(0.02152)	(0.01498)	(0.01460)
MVD	0.00006^{*}	0.00006^{**}	0.00005^{**}
	(0.00003)	(0.00002)	(0.00002)
$MVD \times Post$	0.00007^{**}	0.00006^{**}	0.00005^{**}
	(0.00002)	(0.00002)	(0.00002)
N	7605	32922	61876
Focal/Customers/Suppliers FE	525	1351	2176
R^2	0.80241	0.78693	0.73754
Adj. R^2	0.78770	0.77780	0.72796

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$

EC.8. Cost of Debt Analyses

In quarter q, the cost of debt for a firm is defined as $r_D = XINT_q/[(DLTT_q + DLC_q + DLTT_{q-1} + DLC_{q-1})/2]$ respectively. The metrics can be computed from Compustat Fundamentals Quarterly data items: $DLTT_q$ for total long-term debt, DLC_q for debt in current liabilities, and $XINT_q$ for total interest and related expenses.

We conduct a parallel trend analysis by replacing $Post_{iq}$ in Equation (8) with the dummy variable $\mathcal{I}_{\{QN_{iq}=k\}}$. QN_{iq} ranges from -4 to 4, representing the number quarters relative to the disruption announcement quarter. $\mathcal{I}_{\{QN_{iq}=k\}} = 1$ when $QN_{iq} = k$ and we set β_0 to 0:

$$r_{Diq} = \alpha_i + QN_{iq} + \sum_{k=-4}^{4} \beta_k Long\_RL_{iq} \times \mathcal{I}_{\{QN_{iq}=k\}} + \varepsilon_i. \tag{EC.2}$$

Table EC.5 Panel 1 presents the parallel trend estimation results and show no significant differences in pre-event trends for the treated and control groups.

Additionally, we conduct the falsification test where we randomly reassign the resilience indicator, e.g. $Long\_RL$, across firms while preserving the original proportion of treated firms. Repeating this procedure 1000 times, we generate a distribution of placebo coefficients for each interaction term. As shown in Table EC.5 Panel 3, the true estimates from the baseline regressions are clearly distinct from the distribution of placebo coefficients: the mean placebo estimates are close to zero. The 95% confidence intervals of the placebo coefficients cover zero and exclude the true estimate. Moreover, in more than 90% of placebo iterations, the interaction coefficients are statistically insignificant (%

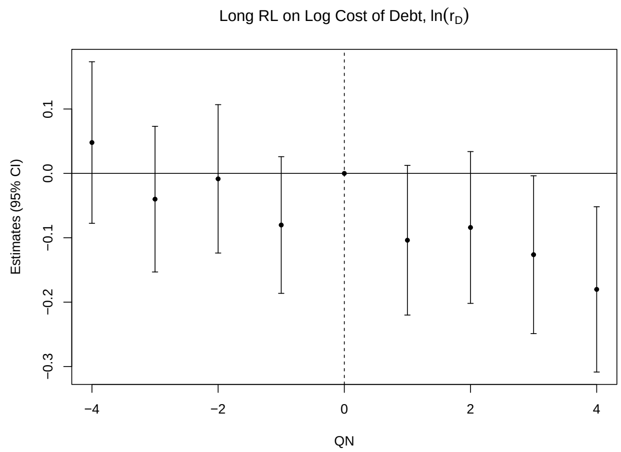
Table EC.5 Parallel Trend and Falsification Test of Cost of Debt (r_D) for Disrupted Firms

Panel 1: Estimates of Parallel Trend

$Long\_RL \times \mathcal{I}_{\{QN=-4\}}$	0.048
	(0.064)
$Long\_RL \times \mathcal{I}_{\{QN=-3\}}$	-0.040
	(0.057)
$Long\_RL \times \mathcal{I}_{\{QN=-2\}}$	-0.008
	(0.059)
$Long\_RL \times \mathcal{I}_{\{QN=-1\}}$	-0.080
	(0.054)
$Long\_RL \times \mathcal{I}_{\{QN=1\}}$	-0.104^{+}
	(0.059)
$Long\_RL \times \mathcal{I}_{\{QN=2\}}$	-0.084
	(0.060)
$Long\_RL \times \mathcal{I}_{\{QN=3\}}$	-0.126^{*}
	(0.062)
$Long\_RL \times \mathcal{I}_{\{QN=4\}}$	-0.180^{**}
	(0.065)
N	3836
Firm FE	287
Relative Quarter FE	9
R^2	0.759

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{+}p < 0.1$

Panel 2: Parallel Trend Plot



Panel 3: Falsification Test

Specification	True Est	Placebo Mean	SE	95% CI	% Null
$\ln r_D \sim Long\_RL \times Post$	-0.084	-0.001	0.026	-0.051 - 0.049	0.902

Null, $p \geq 0.1$). Together, these results suggest that the observed post-disruption interaction effects are unlikely to be driven by chance or by generic post-period dynamics, supporting the robustness of our findings.